

HEARTFIRST BY SHYNTESY · EVIDENCE SYNTHESIS AND
POLICY REVIEW

From Routine to Baseline: Cholesterol Testing

The History, Global Reality, Value,
Limits, and Future of Standard
Cholesterol Testing

A Comprehensive Evidence Review for Preventive
Cardiologists, Lipidologists, Health Researchers, Health
Economists, Policy Makers, and Interested Parties



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Key Findings: Cholesterol Testing Is Foundational, Not Complete

One of prevention's major successes

Routine cholesterol testing enabled the statin era and remains central to cardiovascular prevention worldwide.

LDL-C lowering is outcome-proven

Randomised trials show that each 1 mmol/L LDL-C reduction lowers major vascular events by approximately 22%.

"Routine" is not globally routine

In low- and middle-income countries, only about 40% of adults meeting WHO testing criteria have ever had their cholesterol measured.

A "normal" LDL-C may not be the full picture

ApoB discordance, elevated Lipoprotein(a) (Lp(a)), metabolic risk, inflammatory risk, family history, or subclinical plaque may change interpretation.

"Baseline" is more accurate

The standard lipid panel is best understood as a foundational baseline assessment, not a complete cardiovascular risk evaluation.

Layered assessment is the future

Modern prevention should combine baseline lipids with risk context and selective escalation when evidence, risk, and resources justify it.

Lp(a) once, ApoB selectively

Guidelines increasingly support once-in-adulthood Lp(a) measurement, while ApoB can add precision for high-risk or discordant profiles.

Different settings need different priorities

Expand basic access where testing is absent; improve interpretation and layered assessment where baseline testing is already available.

The standard lipid panel should not be discarded.
It should be understood as the first layer of cardiovascular risk assessment.
Essential, proven, and actionable, but not always sufficient on its own.

What This Means in Practice

The standard panel still matters. The key is to use it well: as a baseline, in context, and with selective escalation when the story does not fit the panel alone.

Do not discard the standard panel

It remains scalable, familiar, inexpensive, and directly connected to proven treatment decisions.

Do not stop when risk seems discordant

If family history, metabolic risk, or clinical context do not match the panel, further assessment may be justified.

Repeat testing should have a purpose

Retesting is most useful when it informs treatment response, adherence, changing risk, or a specific decision.

Results should lead to action

Testing improves prevention only when results are interpreted, explained, and linked to clear next actions.

Use the standard lipid panel as a starting point, not a stopping point.
Context matters. Follow-through matters.
And in selected cases, a fuller risk picture matters too.

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Evidence Summary

Cholesterol testing has journeyed from specialist biochemistry laboratories to the GP consultation room, the workplace health check, and the community pharmacy. In its most common form, a non-fasting or fasting lipid panel measuring total cholesterol, LDL-C, HDL-C, triglycerides, and calculated non-HDL-C has underpinned one of the great prevention success stories of modern medicine. Randomised trials demonstrate that each 1 mmol/L reduction in LDL-C lowers major vascular events by approximately 22%, and Mendelian randomisation studies confirm that LDL-C is causally related to atherosclerotic cardiovascular disease, with lifelong lower levels conferring even greater protection.

Yet the landscape has evolved. The same panel that was revolutionary when the Friedewald equation made LDL-C estimation widely available now reveals its boundaries. LDL-C and apolipoprotein B can diverge meaningfully, especially in insulin-resistant states, and a *normal* LDL-C may mask a high number of atherogenic particles. Lipoprotein(a), an independent, genetically determined risk factor affecting approximately one in five people globally, is not captured by any standard panel. Meanwhile, the word **routine** conceals stark global variation: in low- and middle-income countries, only 39.7% of adults meeting WHO criteria have ever had their cholesterol measured, while in high-income settings over-testing of stable low-risk individuals coexists with under-testing of high-risk groups, including those with familial hypercholesterolaemia.

This evidence review evaluates the proposition that routine cholesterol testing remains historically vital and clinically valuable, but in many contexts is best understood as a **foundational baseline tool rather than a complete modern cardiovascular risk assessment**. The evidence supports that framing in high-resource settings where enhanced assessment is feasible. However, the same language must be applied with care so that it does not inadvertently imply the standard panel is insufficient in settings where even basic lipid testing is not yet reliably available.

The report finds that cholesterol testing is foundational but incomplete. It remains the indispensable first layer of cardiovascular risk evaluation, low-cost, scalable, and proven. But modern prevention, where resources permit, demands a layered approach: baseline lipids plus blood pressure, glycaemic status, smoking history, family history, and renal function, with selective addition of apolipoprotein B, lipoprotein(a), and, in intermediate-risk individuals, coronary artery calcium scoring. Terminology such as **baseline lipid assessment** or **foundational lipid evaluation** better communicates this starting-point role than "standard cholesterol test", which may inadvertently suggest sufficiency and contribute to false reassurance.

The report concludes with five unanswered questions that deserve greater international attention:

1. What is the optimal screening interval for baseline lipid assessment across different age, sex, and risk strata?
2. How cost-effective is universal once-in-lifetime Lp(a) measurement compared with targeted screening in diverse global settings?
3. Would treating to apoB targets rather than LDL-C targets further improve cardiovascular outcomes?
4. How can cholesterol testing be sustainably integrated into primary care in low-resource settings that lack conventional laboratory infrastructure?
5. Does re-framing the test as a **baseline** rather than *standard* assessment alter clinician ordering patterns and patient understanding in ways that reduce both over-testing and under-testing?

Routine cholesterol testing is neither obsolete nor sufficient.

It provides essential baseline information for cardiovascular diagnosis and for guiding treatment decisions that have saved lives. But it is no longer sufficient on its own.

Today's understanding of atherogenesis recognises that risk may be better captured by markers such as apolipoprotein B, which provides a closer estimate of atherogenic particle number because each major atherogenic particle carries one apoB molecule, and lipoprotein(a), an independent genetic contributor to risk that remains hidden in a standard panel.

In this light, routine cholesterol testing is most useful when understood as a proven, widely available, and essential foundation, the first indispensable layer upon which a fuller, stratified cardiovascular prevention strategy can be built, rather than a complete assessment in itself.

1. Definitions and Terminology

1.1 What Is Commonly Meant by "Routine Cholesterol Testing"?

In clinical practice, the terms **routine cholesterol test**, **standard lipid panel**, **lipid profile**, and **cholesterol test** are often used interchangeably in everyday clinical and public-facing language, although they are not always technically identical. A standard lipid panel typically includes the direct measurement of total cholesterol (TC), high-density lipoprotein cholesterol (HDL-C), and triglycerides (TG), with low-density lipoprotein cholesterol (LDL-C) most often calculated rather than measured directly. Non-HDL-C, derived by subtracting HDL-C from TC, is increasingly reported as an additional parameter. The American Heart Association describes a standard cholesterol test as reporting LDL, HDL, total cholesterol, and triglycerides. The UK's NICE defines a full lipid profile as the measurement of TC, HDL-C, and TG, with LDL-C and non-HDL-C calculated from these values. Commercial laboratory catalogues, such as that of Quest Diagnostics, list a standard lipid panel comprising TC, HDL-C, TG, calculated LDL-C, non-HDL-C, and the cholesterol/HDL-C ratio.

Each analyte carries specific physiological meaning:

- **Total cholesterol (TC)** is the aggregate concentration of cholesterol carried in all lipoprotein fractions. TC alone is an imprecise risk marker because it combines atherogenic and anti-atherogenic fractions.
- **LDL-C** is the primary target of lipid-lowering therapy and the cholesterol mass within low-density lipoprotein particles. In routine practice it is most often estimated using the Friedewald equation ($\text{LDL-C} = \text{TC} - \text{HDL-C} - \text{TG}/5$ in mg/dL, or $\text{TG}/2.2$ in mmol/L). The equation becomes unreliable when TG exceed approximately 4.0–4.5 mmol/L and systematically underestimates LDL-C at very low concentrations. Alternative methods, such as the Martin-Hopkins equation or direct homogeneous assays, are available but not universally adopted.
- **HDL-C** is measured directly and functions as an inverse risk marker in most risk calculators. Mendelian randomisation studies suggest HDL-C is better understood as a risk marker than a therapeutic target.
- **Triglycerides (TG)** reflect the concentration of triglyceride-rich lipoproteins, predominantly VLDL, and their remnants. Elevated TG is a marker of triglyceride-rich lipoprotein and remnant-lipoprotein burden and is associated with ASCVD risk; whether TG itself is causal is less clear than the evidence for apoB-containing particle burden.
- **Non-HDL-C** is calculated as TC minus HDL-C and captures the cholesterol content of the major apoB-containing atherogenic particles, including LDL, IDL, VLDL remnants, and Lp(a). (At typical Lp(a) concentrations its contribution to non-HDL-C is minor; it becomes clinically meaningful only at markedly elevated levels, which is one reason Lp(a) is discussed later as a hidden risk factor not adequately captured by the standard panel.)

Non-HDL-C performs at least as well as LDL-C as a risk predictor in most epidemiological studies and better than LDL-C in individuals with hypertriglyceridaemia. Major guidelines now recommend non-HDL-C as a secondary treatment target.

1.2 Fasting Versus Non-Fasting Testing

Historically, lipid panels were performed after a 9- to 12-hour fast. Postprandial TG elevation was thought to confound interpretation, and the Friedewald equation was validated on fasting samples. Contemporary evidence, including large analyses from the Copenhagen General Population Study, has established that non-fasting lipid profiles have prognostic value equivalent to fasting profiles for cardiovascular risk assessment. TC and HDL-C vary minimally after eating, and non-fasting TG values provide additional information about postprandial lipoprotein metabolism that may be independently relevant to risk.

The 2018 ACC/AHA guideline states that non-fasting samples may be used for risk assessment in primary prevention and for establishing baseline LDL-C before initiating a statin. The ESC/EAS 2019 guidelines endorse non-fasting testing as the default for cardiovascular screening, reserving fasting measurements for specific circumstances such as suspected genetic hyperlipidaemia, very high TG (over 4.5 mmol/L), or when precise LDL-C calculation is required. NICE similarly notes that a fasting sample is not mandated for a full lipid profile. This pragmatic shift has improved the accessibility and convenience of lipid testing, particularly in primary care and community settings.

1.3 Variation Across Countries, Guidelines, and Settings

Despite broad agreement on the core analytes, definitions diverge along several axes:

- **Country:** In the United Kingdom, NICE defines a full lipid profile as TC, HDL-C, and TG, with LDL-C and non-HDL-C calculated; the LDL-C calculation method may vary by local laboratory. In the United States, the standard panel reports TC, HDL-C, LDL-C (calculated or direct), and TG, with some laboratories also providing non-HDL-C and the cholesterol/HDL-C ratio. Canadian guidelines recommend non-fasting testing for screening and advise that non-HDL-C or apoB be used for risk stratification when TG exceed 1.5 mmol/L.
- **Guideline:** ESC/EAS 2019 designates LDL-C as the primary treatment target but recommends non-HDL-C and apoB as secondary targets, particularly in individuals with diabetes, high TG, or obesity. ACC/AHA 2018 focuses on LDL-C while acknowledging non-HDL-C and apoB as risk-enhancing factors. NICE NG238 (2023) employs QRISK3, which incorporates TC and HDL-C, for risk estimation. The 2016 USPSTF lipid-screening statement emphasised TC and HDL-C and found insufficient evidence to recommend for or against routine triglyceride measurement; more recent US prevention guidance embeds lipid assessment within broader ASCVD risk-based statin decision-making rather than offering a detailed laboratory-panel definition.

- **Laboratory and payer system:** In the United States, CPT code 80061 for a lipid panel typically includes TC, HDL-C, TG, and calculated LDL-C. Coverage frequency varies: some insurers reimburse annually for high-risk individuals but biennially or less often for low-risk screening. In Australia, Medicare item 699 covers cardiovascular risk assessment, but the exact panel composition may vary by pathology provider.

Summary Table: Terminology Across Major Bodies

Term	Typical Analytes	Context	Guideline or Body
Standard lipid panel	TC, LDL-C, HDL-C, TG, non-HDL-C	Primary care, risk screening	ACC/AHA, ESC/EAS
Lipid profile / Full lipid profile	TC, HDL-C, TG, calculated LDL-C and non-HDL-C	General clinical use	NICE, NHS
Fasting lipid panel	As above, after 9–12 h fast	When TG precision or FH suspected	Historical standard
Non-fasting lipid profile	As above, no fasting	Cardiovascular screening	ESC/EAS 2019, ACC/AHA
Total CVD risk assessment	Lipids plus BP, glucose, smoking status	Primary prevention	WHO/ISH, ESC
Cholesterol test (public)	TC, or full panel depending on context	Consumer, community, and public-facing settings	Variable

1.4 Absence of a Universal Authoritative Definition

No global body has promulgated a universally mandated definition of a "standard cholesterol test". The WHO includes cholesterol measurement in its essential in-vitro diagnostics list without functioning as a universal clinical definition of a standard cholesterol test. The WHO/ISH cardiovascular risk prediction charts exist in two formats: one requiring TC measurement, and another using BMI as a proxy when cholesterol testing is unavailable.

The ACC/AHA 2018 guideline recommends "a standard non-fasting or fasting lipid profile to document baseline lipid levels, estimate ASCVD risk, and guide initiation of lipid-lowering therapy" but does not rigidly define the panel beyond TC, LDL-C, HDL-C, and TG. The ESC/EAS 2019 guidelines refer to a "standard serum lipid profile" comprising TC, HDL-C, and TG, from which LDL-C is estimated. NICE NG238 explicitly defines a full lipid profile as TC, HDL-C, and TG, with LDL-C and non-HDL-C calculated.

The most consistent finding across all authorities is the inclusion of TC and HDL-C. The inclusion of TG and the method by which LDL-C is determined remain the principal sources of variability. This absence of a single definition is not necessarily a weakness: it reflects the reality that lipid testing serves multiple purposes across different clinical contexts and health-system capabilities.

2. Historical Development

2.1 Timeline of Key Milestones

Year	Milestone	Significance
1769 – 1816	Poullietier de la Salle isolates cholesterol from gallstones; Chevreul names it “cholesterine”	Chemical identification of cholesterol
1843	Vogel detects cholesterol in atherosclerotic plaques	First link between cholesterol and arterial disease
1910	Windaus quantifies cholesterol in aortic tissue, finding ~20-fold enrichment in atheromas	Quantitative pathological evidence
1913	Anitschkow demonstrates cholesterol-fed rabbits develop arterial lesions resembling human atherosclerosis	Experimental origin of the lipid hypothesis
1934	Schoenheimer and Sperry develop an early quantitative assay for serum cholesterol	First reproducible clinical measurement
1948	Framingham Heart Study begins enrolling participants	Foundation of prospective cardiovascular epidemiology
1952	Abell, Levy, Brodie, and Kendall publish a simplified, reliable cholesterol assay	Enables wider clinical and research use
1949 – 1955	Gofman and colleagues use ultracentrifugation to separate lipoprotein classes; LDL and HDL identified	Lipoprotein fractionation becomes possible; LDL and HDL concepts enter cardiovascular research
1957	Seven Countries Study initiated (Keys)	Cross-cultural ecological evidence linking diet, cholesterol, and CHD
1958	CDC standardises cholesterol measurement using the Abell–Kendall reference method	Analytical precision achieved; basis for population screening
1961	Early Framingham reports link total cholesterol to incident coronary heart disease, including myocardial infarction and angina pectoris	Cholesterol established as an independent, graded risk factor
1966	Fredrickson, Levy, and Lees publish lipoprotein phenotype classification	Clinical classification of dyslipidaemias
1972	Friedewald equation published ($LDL-C = TC - HDL-C - TG/5$)	LDL-C estimation becomes feasible without ultracentrifugation; one of the most important technical enablers of mass LDL-C reporting
1973 – 1976	Goldstein and Brown characterise the LDL receptor (Nobel Prize 1985)	Molecular mechanism linking LDL to atherosclerosis; legitimises therapeutic targeting
1984	Lipid Research Clinics–Coronary Primary Prevention Trial (LRC-CPPT): cholestyramine reduces CHD events	First large RCT demonstrating cholesterol lowering reduces coronary events
1987	Lovastatin approved by FDA; first statin becomes available	Pharmacological LDL-C lowering enters routine practice

Year	Milestone	Significance
1988	US National Cholesterol Education Program (NCEP) Adult Treatment Panel I launched; mass screening promoted	Cholesterol testing formally recommended for all adults ≥ 20 years
1994	Scandinavian Simvastatin Survival Study (4S): simvastatin reduces total mortality in secondary prevention	Statin mortality benefit proven
1995 – 1996	WOSCOPS and CARE trials extend statin benefit to primary prevention and post-MI populations	Primary prevention evidence consolidated
2001	NCEP ATP III: risk-stratified LDL-C targets; systematic risk calculator use endorsed	Global risk assessment becomes standard; testing volume accelerates
2005	Cholesterol Treatment Trialists' (CTT) Collaboration publishes major individual-participant statin meta-analysis	Establishes proportional reduction in major vascular events per 1 mmol/L LDL-C reduction
2009	NHS Health Check programme introduced in England	Embeds cholesterol measurement in structured population cardiovascular risk assessment for adults aged 40–74
2010	CTT Collaboration evaluates more intensive LDL-C lowering across 26 trials	Reinforces “lower LDL-C, lower risk” dose-response
2013	ACC/AHA guideline shifts to risk-based statin allocation; Pooled Cohort Equations introduced	LDL-C targets de-emphasised; 10-year ASCVD risk drives treatment decisions
2017 – 2018	FOURIER and ODYSSEY OUTCOMES demonstrate PCSK9 inhibitor outcome benefits on top of statin therapy	Very low LDL-C levels shown to be safe and incrementally beneficial
2021 – 2025	EAS consensus statements and ESC/EAS focused updates expand attention to apoB, non-HDL-C, Lp(a), CAC, and risk modifiers	Modern lipid assessment becomes increasingly layered and risk-stratified

2.2 From Biochemical Measurement to Population Prevention

Cholesterol entered clinical medicine slowly. Early colorimetric methods such as the Liebermann–Burchard reaction (1885–1889) required hazardous reagents and organic solvent extraction; they were confined to research laboratories. The development of the Abell–Kendall assay in 1952 provided the first reliable, reproducible method suitable for wider use, and its adoption as the CDC reference method in 1958 gave the field an analytical anchor. Automated enzymatic assays, introduced in the late 1970s, reduced cost and variability sufficiently to make population-level screening operationally feasible.

Epidemiology provided the clinical rationale. The Framingham Heart Study, initiated in 1948, was the first to track serum cholesterol prospectively alongside cardiovascular outcomes in a general population. By 1961, Framingham had established total cholesterol as a graded, independent predictor of incident coronary heart disease, including myocardial infarction and angina pectoris. The Seven Countries Study (1958 onward) extended this finding ecologically, showing that populations with higher mean cholesterol levels experienced

substantially higher CHD mortality, though subsequent analyses have refined the relationship and critiqued its original focus on dietary fat to the exclusion of other nutritional and lifestyle variables. The INTERHEART study (2004) demonstrated that a small set of modifiable risk factors accounted for most first myocardial infarction risk across 52 countries, underscoring the global relevance of lipid-associated risk.

The mechanistic breakthrough came from Goldstein and Brown, who characterised the LDL receptor in the 1970s. Their work explained why familial hypercholesterolaemia produced catastrophic premature atherosclerosis: defective receptor-mediated clearance of LDL particles led to markedly elevated plasma LDL-C. This transformed cholesterol from a statistical risk correlate into a biologically understood driver of disease. The Friedewald equation, published in 1972, made LDL-C estimation possible without preparative ultracentrifugation. By allowing any clinical laboratory to report LDL-C from the standard panel of TC, HDL-C, and TG, the equation was one of the most important technical enablers of mass LDL-C reporting.

Even as this LDL-C-centric paradigm consolidated, epidemiological evidence was beginning to highlight the importance of particle number over cholesterol mass. A subsequent INTERHEART lipid analysis (2008) found that the apoB/apoA1 ratio outperformed conventional cholesterol ratios as a global marker of myocardial infarction risk, an early signal of the limitations that would later drive interest in apolipoprotein B and non-HDL-C.

The demonstration that lowering cholesterol reduces clinical events arrived in stages. The Lipid Research Clinics–Coronary Primary Prevention Trial (1984) showed that the bile acid sequestrant cholestyramine reduced CHD events, but the effect size was modest and the drug poorly tolerated. The statin era changed everything. 4S (1994) demonstrated that simvastatin reduced total mortality in patients with established CHD. WOSCOPS (1995) extended this benefit to men with elevated cholesterol and no prior myocardial infarction. CARE and LIPID confirmed and extended these findings. The CTT Collaboration subsequently showed, across more than twenty randomised trials, that each 1 mmol/L reduction in LDL-C lowers major vascular events by approximately 22%, with benefit proportional to absolute LDL-C reduction.

This trial evidence catalysed guideline adoption. The NCEP ATP III guidelines (2001) recommended universal lipid screening for adults over 20 and set LDL-C treatment targets stratified by global risk. The 2013 ACC/AHA guideline shifted to a risk-based statin allocation model, embedding lipid measurement within a broader clinical risk calculation rather than treating LDL-C as a standalone treatment trigger. In the UK, the NHS Health Check programme (formally launched 2009) embedded cholesterol measurement within a structured cardiovascular risk assessment offered to all adults aged 40–74 every five years.

2.3 The Shift from Specialist Tool to Routine Primary Care

Cholesterol testing moved from specialist lipid clinics to routine primary care through converging forces. Enzymatic methods compatible with automated clinical chemistry analysers made testing cheap, rapid, and reproducible. The commercial availability of statins created a direct instrumental need for measurement: without a lipid panel, there was no indication, no dose titration, and no monitoring. Practice guidelines from NCEP, ESC, and ACC/AHA systematically recommended population screening. Risk calculators (Framingham, SCORE, QRISK, and later PREVENT and SCORE2) integrated lipid values into actionable clinical estimates. An audit from Oxfordshire, UK (O'Sullivan et al., 2011) documented a more than 15-fold rise in lipid test requests between 1987 and 2007, with testing rising most sharply after the publication of major statin trials, predominantly in primary care settings.

This historical success is inseparable from the history of the evidence-based prevention paradigm itself. Cholesterol testing enabled the identification of high-risk individuals, the deployment of proven therapies, and the monitoring of treatment response at scale. It is a genuine public-health achievement. But the same history that embedded the standard lipid panel in routine care also cemented certain simplifications that are now being re-examined: reliance on calculated LDL-C rather than direct particle measurement, a focus on cholesterol mass rather than particle number, and an implicit framing of the test as a sufficient rather than a starting assessment. Understanding this arc is essential for evaluating whether "baseline lipid assessment" better describes the panel's contemporary role than "standard cholesterol test".

3. Purpose and Clinical Decisions

Routine lipid panels are among the most versatile tools in cardiovascular medicine. They support decisions across primary prevention, dyslipidaemia diagnosis, treatment initiation, treatment intensification, therapeutic monitoring, familial risk detection, and population-level risk stratification. Yet the distinction between their intended uses and their real-world deployment reveals both the strength of the standard panel and the ways it can be misused when separated from clinical context.

3.1 Intended Uses

Primary prevention risk estimation. The most common intended use in primary care is the integration of lipid values into multivariable risk calculators. Total cholesterol and HDL-C are core lipid inputs for many laboratory-based risk tools, including SCORE2, the Pooled Cohort Equations, PREVENT, QRISK3, and Framingham-derived models. LDL-C is not always required for the risk calculation itself, but it is central to treatment thresholds, treatment intensity, and follow-up monitoring. Without lipid measurement, most laboratory-based calculators cannot be applied, although WHO and other global-health approaches provide non-laboratory alternatives when cholesterol testing is unavailable.

Diagnosis of dyslipidaemia. The standard panel provides the first diagnostic layer for conditions ranging from mild hypercholesterolaemia to severe combined hyperlipidaemia and hypertriglyceridaemia. It is also the initial screen for familial hypercholesterolaemia (FH), an autosomal dominant condition affecting approximately 1 in 250 people worldwide, characterised by markedly elevated LDL-C from birth and a correspondingly high lifetime cardiovascular risk if untreated. An LDL-C above 4.9 mmol/L (190 mg/dL) in adults, or elevated values in younger individuals with a supportive family history, triggers FH suspicion, specialist referral, and cascade family testing. Similarly, very high triglycerides (≥ 10 mmol/L) prompt investigation for secondary causes and pancreatitis risk.

Statin eligibility and treatment intensity. In most guideline systems, LDL-C thresholds and risk-based targets directly determine who should receive lipid-lowering therapy and at what intensity. The ESC/EAS 2019 guidelines set LDL-C goals of <1.4 mmol/L for very high-risk patients and <1.8 mmol/L for high-risk patients, while the ACC/AHA 2018 guideline uses percentage reduction targets. Both approaches require a baseline LDL-C measurement. In the United Kingdom, NICE recommends statin therapy when 10-year cardiovascular risk, calculated using TC and HDL-C, reaches or exceeds 10%.

Lifestyle counselling. Lipid results provide a concrete, quantifiable metric that can motivate dietary modification, physical activity, and weight management. Elevated triglycerides may specifically prompt advice on alcohol and carbohydrate intake. A "normal" panel can reassure individuals who are genuinely at low risk, provided other risk factors have been assessed, but it should not deter attention to those factors.

Follow-up monitoring. Serial lipid panels after the initiation of lipid-lowering therapy assess adherence, therapeutic response, and the need for dose adjustment or additional agents. Most guidelines recommend re-checking a lipid profile about 6–12 weeks after starting or changing lipid-lowering therapy, including statin therapy, and periodically thereafter. This monitoring function is often the most frequent indication for repeat testing.

Secondary prevention. In patients with established atherosclerotic cardiovascular disease, lipid panels confirm very high-risk status and guide intensive therapy. Typical LDL-C targets are <1.4 mmol/L (55 mg/dL), and failure to reach them prompts escalation to ezetimibe or PCSK9 inhibitors. Lipid testing is recommended during hospitalisation for acute coronary syndromes and at follow-up.

Diabetes and metabolic risk assessment. Dyslipidaemia is a core component of the metabolic syndrome and diabetic dyslipidaemia. Lipid testing is recommended at diagnosis of diabetes and periodically thereafter, even when LDL-C appears unremarkable, because non-HDL-C or apoB may be elevated and statin therapy is often indicated on the basis of diabetes alone as a risk enhancer.

Population-level screening and risk stratification. National programmes such as the NHS Health Check in England embed cholesterol measurement within structured cardiovascular risk assessment for adults aged 40–74 every five years. The World Health Organization's HEARTS technical package includes lipid measurement where feasible, while also supporting non-laboratory risk charts that use variables such as age, sex, blood pressure, smoking status, diabetes, and BMI when cholesterol testing is unavailable. In several emerging economies, guidelines are shifting toward earlier or broader testing: Chinese lipid guidelines recommend lipid testing as a routine item in primary, middle, and high school physical examinations, while South African guidance recommends at least one lipid assessment from age 20, partly reflecting high familial hypercholesterolaemia prevalence in some communities.

3.2 Real-World Uses

Over-testing. The Oxfordshire audit documented a more than 15-fold increase in lipid test requests between 1987 and 2007. Repeat monitoring, much of it without clear evidence of changing management, grew from 24% of all lipid tests in 1993–1995 to 61% in 2005–2007, with an estimated 42–79% of tests in the latter period being potentially unnecessary. Among patients with established coronary heart disease who had already achieved LDL-C targets, roughly one-third of repeat panels were performed without any subsequent intensification of therapy. These patterns suggest that lipid panels are often used as a habitual general-health check rather than as a targeted component of risk-based prevention.

However, over-testing and under-testing frequently coexist within the same systems.

Under-testing is prevalent among populations at elevated risk. In one multi-country analysis of low- and middle-income countries, among adults meeting WHO Package of Essential Noncommunicable Disease Interventions (WHO PEN) criteria for diagnostic cholesterol

testing, only 39.7% reported ever having had their cholesterol measured. Even in high-income settings, screening uptake varies substantially by socioeconomic status, ethnicity, and geography. Young adults with familial hypercholesterolaemia, who stand to gain the most from early detection, remain disproportionately unscreened in many systems.

Isolation from full risk assessment. In routine primary-care workflows, lipid results may be reviewed without consistent simultaneous blood pressure assessment, formal risk calculator use, or structured lifestyle counselling. In these instances, the test's potential value is not fully realised, not because the panel itself is inadequate but because the surrounding clinical infrastructure has not been optimised to deliver integrated cardiovascular prevention.

Misinterpretation. A "normal" lipid panel can provide false reassurance when LDL-C is within the population reference range but elevated relative to the individual's risk-based target, when apoB is discordantly high, or when non-lipid risk factors such as smoking or hypertension dominate the overall risk profile. Conversely, minor borderline elevations in low-risk individuals may trigger unnecessary anxiety and repeat testing.

The contrast between intended and real-world use underscores a central tension. The same panel that is indispensable for risk estimation, diagnosis, treatment initiation, and monitoring can, when deployed without context, become a source of therapeutic inertia, false reassurance, anxiety, or wasted resource. This duality is central to the argument developed in this report: reframing the lipid panel as a baseline or foundational assessment may help preserve its value while making clearer that it is not, by itself, a complete cardiovascular risk evaluation.

4. Strengths and Achievements

The value of routine cholesterol testing must be weighed in full before any critique is offered. The history and evidence summarised in this section represent one of the most rigorous and high-impact translational achievements in modern medicine.

4.1 Evidence Linking LDL-C and ApoB-Containing Lipoproteins to ASCVD

The causal role of LDL-C and other apoB-containing lipoproteins in atherosclerotic cardiovascular disease (ASCVD) is supported by multiple converging lines of evidence that are rarely matched for any other biomarker.

Randomised trial evidence. The Cholesterol Treatment Trialists' (CTT) Collaboration meta-analyses of individual-participant data from over 170 000 participants in 26 randomised trials demonstrate that each 1.0 mmol/L reduction in LDL-C reduces major vascular events by approximately 22 % (RR 0.78, 95 % CI 0.76–0.80), with proportional reductions across age, sex, and baseline risk categories. Subsequent non-statin trials (IMPROVE-IT, FOURIER, ODYSSEY OUTCOMES, CLEAR Outcomes) have confirmed that LDL-C lowering with ezetimibe, PCSK9 inhibitors, and bempedoic acid further reduces events, making it unlikely that the observed benefit is purely a class-specific statin effect. The consistency of benefit across mechanistically distinct drug classes strengthens causal inference beyond what any single trial design could provide.

Mendelian randomisation evidence. Genetic variants that mimic lifelong lower LDL-C exposure confer substantially greater protection than short-term pharmacological lowering. Each 1 mmol/L genetically lower LDL-C is associated with an approximately 55 % reduction in coronary heart disease risk, compared with roughly 22 % for statin-mediated lowering over approximately five years. This gene-dose relationship is observed for variants in *PCSK9*, *HMGCR* (the statin target), *NPC1L1* (the ezetimibe target), and *LDLR*, collectively establishing that cumulative LDL-C exposure, not a particular drug mechanism, drives risk. Trial and genetic evidence has not identified a clear lower LDL-C threshold below which benefit reliably disappears in appropriately selected high-risk patients, although absolute benefit depends strongly on baseline risk: the absolute reduction in events is determined primarily by a patient's underlying cardiovascular risk, not by LDL-C level alone.

Observational evidence. Prospective cohort studies, including Framingham, the Seven Countries Study, the Emerging Risk Factors Collaboration, and the Global Burden of Disease investigation, have demonstrated strong, graded, and independent associations between LDL-C (and non-HDL-C) and ASCVD events. The INTERHEART case-control study, conducted across 52 countries, estimated that a raised apoB/apoA-I ratio accounted for about half of the population-attributable risk of first myocardial infarction, with adjusted estimates around 49 %, more than any other single modifiable risk factor in that analysis.

4.2 Role of Cholesterol Testing in Enabling Treatment

Cholesterol testing is the gateway to lipid-lowering therapy. Without measurement, neither diagnosis nor treatment monitoring is possible. The global deployment of statins, one of the most cost-effective interventions in modern medicine, rests entirely on the availability of cholesterol testing.

High-intensity statins reduce LDL-C by 50–55 %; adding ezetimibe produces an additional ~20 % relative reduction; PCSK9 monoclonal antibodies (evolocumab, alirocumab) lower LDL-C by approximately 50–60 % and have demonstrated cardiovascular outcome benefits. Inclisiran lowers LDL-C by approximately 50 % with twice-yearly maintenance dosing, with cardiovascular outcome trial data awaited. Every one of these therapies requires LDL-C measurement to establish baseline, titrate dosing, confirm efficacy, and satisfy regulatory and reimbursement requirements. In this sense, the lipid panel is not merely a diagnostic test but the biomarker infrastructure that makes an entire therapeutic ecosystem operable.

The clinical utility of the standard panel extends beyond the management of ASCVD risk. Lipid testing also enables the identification of individuals with extreme elevations requiring urgent intervention: familial hypercholesterolaemia (FH) is unmasked by LDL-C values above 4.9 mmol/L (190 mg/dL), and triglyceride levels above 10 mmol/L alert clinicians to pancreatitis risk independent of ASCVD.

4.3 Public-Health Value and Cost-Effectiveness

Lipid testing is inexpensive, scalable, and can be performed on widely available automated analysers. The incremental cost of adding a lipid panel to a routine clinical encounter is generally low, particularly relative to downstream costs of cardiovascular events or more advanced risk-stratification tools. Combined with generic statin therapy, lipid-testing-guided prevention generates one of the most favourable cost-effectiveness profiles in healthcare.

Economic analyses have consistently found that statin therapy for primary prevention in intermediate- to high-risk individuals is cost-effective, and in some analyses cost-saving, depending on the population targeted and the healthcare setting. A 2015 modelling study by Pandya and colleagues estimated that statin initiation at a 10-year ASCVD risk threshold of $\geq 7.5\%$ had an incremental cost-effectiveness ratio of approximately \$37 000 per QALY compared with a $\geq 10\%$ threshold, well within accepted willingness-to-pay thresholds in the United States. In the United Kingdom, modelling studies have generally estimated the NHS Health Check programme, which embeds cholesterol measurement within cardiovascular risk assessment, to be cost-effective under assumed uptake and treatment scenarios, although real-world effectiveness depends on implementation quality. In high-burden countries, screening high-risk adults and treating those with elevated cholesterol is often cost-saving by preventing expensive acute cardiovascular events.

4.4 Simplicity, Scalability, and Accessibility

The standard lipid panel can be performed on a non-fasting sample (increasing convenience), requires only a venepuncture or finger-stick, and produces results within hours. Point-of-care devices now enable lipid testing in community pharmacies, workplaces, and low-resource settings without conventional laboratory infrastructure, although quality assurance, calibration, and confirmatory laboratory testing remain important when results will be used to determine long-term treatment. The test is standardised internationally through programmes such as the CDC Cholesterol Reference Method Laboratory Network, ensuring comparability across laboratories and over time.

These features make the standard lipid panel uniquely scalable at population level, a quality that more sophisticated risk-stratification tools, such as coronary artery calcium scoring, advanced biomarker panels, and genetic testing, cannot yet match globally. Its simplicity has enabled integration into primary care, pharmacies, workplaces, and national prevention programmes. That scalability is one of the panel's greatest strengths and one reason any modern critique must be careful: the standard lipid panel is not obsolete because it is incomplete; it is valuable precisely because it provides a low-cost, reproducible baseline from which more detailed assessment can proceed where appropriate.

5. Global Variability: How Routine Is "Routine"?

The word "routine" can imply predictable, systematic access. Globally, however, cholesterol testing is far from routine for large parts of the population. Its integration into clinical practice varies by national income, health-system design, reimbursement model, laboratory infrastructure, public-health philosophy, and historical policy choices. In some countries, a lipid panel is a codified element of a free, age-based preventive health check. In others, it is a private-sector commodity, an employer benefit, a tertiary-hospital investigation, an opportunistic primary-care test, or simply unavailable.

This variation matters for both evidence interpretation and policy language. In high-resource systems, the standard lipid panel may be common enough to risk habitual overuse in low-risk individuals. In lower-resource settings, even a single baseline cholesterol measurement may remain inaccessible for many people who would benefit from risk assessment. The term "routine cholesterol testing" therefore describes a highly uneven global reality.

5.1 High-Income Settings

In most high-income countries, lipid testing is widely available, generally reimbursed, and recommended by national or professional guidelines. The precise trigger, frequency, and degree of systematisation nonetheless vary substantially.

United Kingdom. In England, the NHS Health Check programme, introduced in 2009, offers a free cardiovascular risk assessment every five years to adults aged 40–74 without known cardiovascular disease, diabetes, or chronic kidney disease. The assessment includes cholesterol measurement, with cardiovascular risk estimated using QRISK3 and treatment decisions guided by NICE recommendations. Uptake varies considerably by locality, socioeconomic position, ethnicity, and primary-care capacity. Outside the formal Health Check pathway, opportunistic and risk-based lipid testing through general practice remains common. NICE recommends measuring total cholesterol and HDL-C to estimate cardiovascular risk and defines a full lipid profile as total cholesterol, HDL-C, triglycerides, calculated LDL-C, and calculated non-HDL-C, without requiring a fasting sample.

United States. The United States does not have a single national cholesterol-screening programme. Lipid testing is instead delivered through a fragmented mixture of preventive primary care, employer health screening, Medicare wellness visits, private insurance, public programmes, and self-pay laboratory access. Under the Affordable Care Act, many preventive services with USPSTF A or B ratings must be covered by non-grandfathered private health plans without cost-sharing. However, current USPSTF guidance is framed primarily around statin use for primary prevention in adults aged 40–75 with at least one cardiovascular risk factor and sufficient estimated 10-year risk, rather than as a universal lipid-screening mandate. ACC/AHA guidance recommends periodic assessment of traditional risk factors, including dyslipidaemia, beginning in young adulthood. In practice, lipid

screening rates are high among insured and regularly treated populations but incomplete overall. Over-testing of stable low-risk individuals and under-testing of uninsured, underinsured, rural, and marginalised groups can coexist within the same system.

Canada. Canadian guidance recommends lipid or lipoprotein screening for men and women over 40 years of age, or earlier in the presence of specific risk conditions such as diabetes, hypertension, chronic kidney disease, smoking, inflammatory disease, premature family history, or suspected familial hypercholesterolaemia. Non-fasting testing is endorsed for most screening purposes. Repeat cardiovascular risk assessment is generally recommended every five years in adults aged 40–75, with longer intervals sometimes appropriate for clearly low-risk individuals depending on local guidance. Testing is publicly funded, but access and follow-up may be less consistent in remote, rural, and Indigenous communities.

Australia. Australia provides a Medicare-subsidised Heart Health Check for adults aged 45 and over, and from age 30 for Aboriginal and Torres Strait Islander peoples. The assessment includes cholesterol testing alongside blood pressure, diabetes risk assessment, smoking status, and broader cardiovascular risk estimation. Separate clinical guidance commonly recommends lipid testing from age 45, or earlier for Aboriginal and Torres Strait Islander peoples and those with risk factors. Coverage is formally supported through Medicare, but uptake is incomplete. Rural, remote, and Indigenous communities continue to face access barriers, including geographic distance, workforce limitations, and broader structural inequities.

European Union. Most European Union member states provide cholesterol testing through publicly funded or statutory health systems, but the trigger, frequency, and implementation model vary. Some countries use structured health checks. Germany, for example, provides a statutory preventive health check once between ages 18 and 34, and every three years from age 35; from age 35, blood testing includes a lipid profile and glucose measurement. Other European countries rely more heavily on opportunistic testing, occupational health assessment, or risk-based screening in primary care. ESC/EAS guidance recommends systematic cardiovascular risk assessment using SCORE2 or related tools in adults without established ASCVD, diabetes, or chronic kidney disease, with total cholesterol and HDL-C serving as core inputs. National implementation differs substantially by health-system capacity, reimbursement structure, and primary-care organisation.

5.2 Middle-Income Settings

Middle-income countries exhibit substantial heterogeneity. In many, cholesterol testing is expanding but remains unevenly distributed. Urban, affluent, and privately insured populations often have far greater access than rural, low-income, or publicly dependent groups.

India. India does not have a universal national cholesterol-screening programme. Professional society guidelines recommend lipid testing from early adulthood or in the presence of risk factors, but implementation is fragmented. Private laboratories offer lipid

profiles at relatively low cost, and testing is common among urban middle-class and privately insured populations. Public-sector access is more limited and varies substantially by state, facility type, and local health-system capacity. There is no consistently implemented national screening interval. Awareness of cholesterol status remains low relative to the size of the population and the burden of premature cardiovascular disease. India accounts for a disproportionate share of global cardiovascular mortality, and South Asian populations develop atherosclerotic disease at lower LDL-C levels and younger ages than European-derived risk models anticipate, making the gap in routine lipid assessment particularly consequential. Testing is therefore often opportunistic, event-triggered, employer-driven, or privately purchased.

China. Dyslipidaemia prevalence is high in China, but awareness, treatment, and control rates have historically remained low. National survey data have shown that among adults with dyslipidaemia, only a minority are aware of their condition, fewer receive treatment, and fewer still achieve control. Testing is increasingly available in urban hospitals, community health centres, and organised health examinations, but rural access remains more constrained. Recent Chinese lipid guidelines have emphasised earlier detection, including lipid testing as a routine item in primary, middle, and high school entry physical examinations. These recommendations signal a shift toward earlier life-course prevention, although implementation is likely to vary by region, school system, and health-service capacity.

Brazil. Brazil's Unified Health System (SUS) provides access to cholesterol testing, and national cardiovascular prevention guidance recommends lipid assessment in adults with risk factors and within broader risk-stratification pathways. Private-sector testing is also common among insured and higher-income groups. Implementation remains uneven. Public primary-care capacity, regional variation, laboratory availability, and continuity of follow-up affect whether abnormal lipid results translate into treatment. As in many middle-income settings, Brazil has a dual-track pattern: lipid testing may be readily available in private care while more inconsistently implemented in overstretched public-sector clinics.

South Africa. South Africa provides public-sector laboratory testing through the National Health Laboratory Service, and private-sector lipid testing is widely available through laboratories, clinics, and pharmacies. In the public sector, lipid testing is often opportunistic, risk-based, linked to chronic disease management, or performed in the context of competing clinical priorities. South African guidance recommends at least one lipid assessment from age 20, partly reflecting a high local prevalence of familial hypercholesterolaemia in some communities. However, preventive lipid screening competes with substantial burdens from HIV, tuberculosis, diabetes, hypertension, and other health-system pressures. Infrastructure, workforce constraints, and continuity of care limit systematic population-wide implementation.

Gulf States. Several Gulf states, including Qatar, Saudi Arabia, and the United Arab Emirates, have developed or expanded periodic health-screening programmes that include cholesterol testing. These programmes are often publicly funded for citizens and may be delivered

through employer systems, private insurance, or occupational health services for expatriate workers. Dyslipidaemia is a major public-health priority given high regional burdens of obesity, diabetes, and cardiometabolic risk. However, coverage may differ between citizens and expatriates, and follow-up pathways after abnormal results can vary. Because expatriates constitute a majority of the population in several Gulf states, inequities in lipid testing access are not a marginal concern but a population-level one. In this region, cholesterol testing may be increasingly routine in organised health checks but not uniformly integrated into longitudinal primary prevention for all resident populations.

5.3 Low-Income Settings

In low-income countries, cholesterol testing is often opportunistic, facility-based, and concentrated in tertiary hospitals, urban clinics, donor-supported programmes, or private laboratories. For many people, testing is unavailable, unaffordable, or not prioritised within routine primary care.

A cross-sectional analysis of 57 nationally representative surveys from low- and middle-income countries found that among individuals meeting WHO PEN criteria for diagnostic cholesterol testing, only 39.7% had ever had their cholesterol measured. Testing was more common among women, wealthier individuals, and those with higher education, indicating that access is often driven more by sociodemographic advantage than by clinical need.

The WHO HEARTS technical package and WHO PEN guidance promote total cardiovascular risk assessment, but they recognise that laboratory-based testing is not always feasible. Non-laboratory risk charts allow risk estimation using variables such as age, sex, blood pressure, smoking status, diabetes, and body mass index when cholesterol testing is unavailable. In many settings, the absence of reliable laboratory infrastructure, supply chains, trained personnel, affordable medications, and longitudinal primary care makes universal cholesterol testing unrealistic in the short term.

This does not mean cholesterol measurement lacks value in low-resource systems. Rather, it means that testing must be prioritised strategically: for people with established cardiovascular disease, diabetes, hypertension, suspected familial hypercholesterolaemia, strong family history, or other high-risk features. Where testing capacity is limited, the policy question is not whether cholesterol measurement is useful, but how to deploy it without displacing interventions that may offer greater immediate population benefit, such as blood pressure control, tobacco cessation, diabetes care, and affordable statin access.

5.4 Cross-Cutting Observations

Several patterns emerge from the global evidence.

First, "routine" cholesterol testing, as understood in some high-income health systems, systematically offered to defined age groups, reimbursed, and embedded in structured prevention pathways, is the global exception, not the rule.

Second, even within high-income countries, access and follow-up differ by insurance status, socioeconomic position, ethnicity, geography, Indigenous status, and primary-care capacity.

Third, in many middle- and low-income countries, lipid testing often functions as a private-sector commodity or opportunistic facility-based test, creating systematic inequality in who receives cardiovascular risk assessment.

Fourth, where testing is easily available, it may drift toward habitual repeat testing in stable low-risk individuals while high-risk groups remain under-served.

Fifth, guidelines do not implement themselves. Awareness, treatment, and control of dyslipidaemia may remain low even where testing is nominally available.

Sixth, the value of cholesterol testing depends not only on the blood draw but also on the system around it: risk calculation, result interpretation, access to medication, adherence support, follow-up, and capacity to intensify treatment when needed.

These patterns are central to the language question addressed later in this report. Calling the lipid panel a "standard cholesterol test" may carry an unspoken assumption of universal access that does not reflect global reality. At the same time, reframing it as a **baseline** or **foundational** assessment must be done with care. In high-resource settings, that language can help reduce false reassurance and support layered risk assessment. In low-resource settings, however, it should not imply that the standard panel is inadequate or unworthy of investment when even one lipid measurement remains out of reach for many people. The appropriate screening strategy cannot be separated from the health-system context in which it is deployed.

Global Comparison Table

Global Comparison Table: High-Income Systems

Country / Region	Routine?	Typical Trigger	Frequency	Coverage / Cost	Key Constraints
UK (England)	Yes, through NHS Health Check and primary care	Age-based Health Check for adults 40–74 without known CVD, diabetes, or CKD; opportunistic and risk-based testing otherwise	Every 5 years for NHS Health Check	NHS-funded; free at point of care	Incomplete up-take; locality variation; socioeconomic and ethnic disparities
United States	Partial; widely available but fragmented	Risk-based primary care, wellness visits, employer screening, statin-risk assessment	Variable; often periodic in primary care	Insurance-based; \$0 to substantial self-pay depending on insurance and setting	Fragmented system; insurance gaps; inconsistent follow-up; over-testing and under-testing coexist

Country / Region	Routine?	Typical Trigger	Frequency	Coverage / Cost	Key Constraints
Canada	Yes, risk-based and age-based	Men and women over 40; earlier with diabetes, hypertension, CKD, smoking, family history, inflammatory disease, or suspected FH	Often every 5 years for risk assessment; longer intervals may apply in clearly low-risk individuals	Publicly funded; usually free	Remote, rural, and Indigenous access; continuity of care
Australia	Yes, through Medicare-supported risk assessment	Heart Health Check from age 45; from age 30 for Aboriginal and Torres Strait Islander peoples; earlier with risk factors	Claimable every 12 months for eligible Heart Health Check	Medicare-funded; free if bulk-billed, otherwise possible gap payment	Rural and remote access; Indigenous disparities; uptake limitations
Germany	Yes, statutory preventive check-up	Adults 18–34 once; adults ≥35 every 3 years, with lipid profile included from ≥35	Every 3 years from age 35	Statutory insurance; free for insured adults	Participation variability; implementation differences

Global Comparison Table: Middle-Income and Mixed Systems

Country / Region	Routine?	Typical Trigger	Frequency	Coverage / Cost	Key Constraints
India	No universal national programme	Opportunistic, risk-based, employer-based, private laboratory testing	Variable	Mostly private-sector or state-dependent public access; low private prices in urban areas, variable public access	No universal programme; rural access; public-sector capacity; low awareness
China	Partial and expanding	Opportunistic, community health checks, hospital screening, school physical examinations under recent guidance	Variable; periodic by programme and age group	Mixed public, insurance, employer, and private pathways; low in public/community settings, higher in private settings	Urban-rural gap; low awareness, treatment, and control; regional implementation variation
Brazil	Partial	Opportunistic, risk-based, public primary care, private insurance screening	Variable	SUS public system plus private insurance; free in public system, private costs vary	Public-system capacity; regional inequality; poor treatment control in many settings
South Africa	Partial	Opportunistic, risk-based, chronic disease care, private laboratory or pharmacy testing	Variable	Public-sector laboratory services plus private sector; free or subsidised in public care, private costs vary	Resource constraints; competing health priorities; public-private inequality
Gulf States	Increasingly routine in organised health checks	Annual or periodic health checks; citizen programmes; employer or insurance-based screening for expatriates	Annual or periodic depending on programme	Public systems, employer programmes, private insurance; often free for citizens, variable for expatriates	Citizen-expatriate coverage differences; follow-up pathways; evolving primary-care infrastructure

Global Comparison Table: Low-Income and Constrained Settings

Country / Region	Routine?	Typical Trigger	Frequency	Coverage / Cost	Key Constraints
Low-income settings, general	Largely unavailable or opportunistic	Tertiary care, private laboratories, high-risk clinical encounters	Rare or ad hoc	Limited public availability; private self-pay where available; often unaffordable	Laboratory capacity; supply chains; workforce; cost; treatment access; competing health burdens

These disparities have profound implications for the language and policy models that accompany lipid testing. A "standard cholesterol test" is not standard everywhere. A "routine lipid panel" is not routine for many people. A "baseline lipid assessment" may be the most accurate clinical framing in high-resource settings, but globally it also carries an ethical reminder: the first priority in many health systems is still ensuring that baseline testing is available, affordable, interpretable, and linked to effective prevention.

6. WHO and Global Health Perspective

6.1 WHO's Position on Cholesterol Testing

The World Health Organization has not recommended universal, population-based cholesterol screening. Instead, WHO promotes **total cardiovascular risk assessment**, in which cholesterol measurement is one component alongside blood pressure, smoking status, age, sex, and diabetes status. The WHO/ISH risk prediction charts, first published in 2007 and subsequently updated, exist in two versions: one requiring laboratory measurement of total cholesterol, and a simpler version using BMI in place of total cholesterol when laboratory testing is unavailable. Raised total cholesterol remains part of WHO's NCD surveillance architecture; the Global Monitoring Framework includes the age-standardised prevalence of raised total cholesterol and mean total cholesterol concentration as monitoring indicators, though these are surveillance metrics rather than a screening mandate.

The WHO PEN for primary care in low-resource settings recommends targeted diagnostic testing for cardiovascular risk factors—blood pressure, blood glucose, and cholesterol—based on clinical criteria such as BMI above 30 kg/m², or above 25 kg/m² among people aged 40 years and older, rather than universal population-wide cholesterol screening. The WHO HEARTS technical package reinforces this conditional approach, promoting task-sharing, simplified treatment protocols, and the use of non-laboratory risk charts where lipid measurement is not feasible. HEARTS protocols have been implemented in over 30 countries, demonstrating that simplified cardiovascular risk management with conditional lipid testing can be effectively embedded in primary care, lending operational evidence to the policy framework.

6.2 Why Universal Lipid Testing Is Not a Global Priority

Several legitimate policy considerations explain why the WHO does not prioritise universal cholesterol screening, instead recommending a total cardiovascular risk approach to identify those most likely to benefit from treatment. This is especially relevant in regions with a high burden of premature heart attack and stroke, where the opportunity for efficient, targeted prevention is greatest.

Cost and laboratory infrastructure. Reliable enzymatic lipid assays require consistent cold-chain supply, calibrated photometric equipment, quality-controlled reagents, and trained laboratory staff. In many low-income health systems, these conditions cannot be sustainably maintained at the primary-care level. The WHO itself has developed risk charts that can be used without cholesterol measurement, acknowledging this reality. Point-of-care lipid analysers exist and are improving in accuracy and cost, but implementation at scale requires investment and training that competes with other urgent health priorities.

Treatment access. Identifying individuals with elevated cholesterol is of limited value if lipid-lowering therapy is unavailable or unaffordable. Generic statins are now inexpensive and increasingly available through WHO Essential Medicines List inclusion, but distribution infrastructure, prescribing capacity, and consistent supply remain variable across low- and middle-income countries. In settings where statins are not reliably stocked in the public sector, a cholesterol result cannot be linked to effective treatment.

Competing disease burdens. In settings where communicable diseases, maternal and child health, and malnutrition remain dominant priorities, cardiovascular risk screening competes for limited resources. The time horizon for benefit from lipid lowering is longer than for many acute interventions, making it harder to prioritise in short-term health-system planning.

Workforce constraints. The WHO PEN protocol is designed for use by non-physician health workers in primary care. Adding universal cholesterol testing would require phlebotomy skills, equipment maintenance, and supply chains that many primary-care facilities lack.

Risk-based versus universal strategies. The WHO has consistently favoured risk-based approaches over universal screening, on the grounds that treatment should be targeted to those at highest absolute risk. This maximises the number of cardiovascular events prevented per test performed and per unit of healthcare expenditure, while minimising unnecessary medicalisation of low-risk individuals. A risk-based model is particularly efficient in younger populations where event rates are low.

A 2023 study in *The Lancet Global Health* concluded that diagnostic testing availability should be determined by evidence-based criteria and guidelines, not by income or education. The same study found that testing disparities persist and that adherence to WHO PEN testing guidelines is low in many countries.

In summary, WHO treats cholesterol measurement as one tool among several in cardiovascular prevention. It endorses a tiered strategy in which high-risk individuals are prioritised for testing and treatment, population-level lifestyle interventions address risk factors broadly, and risk calculators—with or without cholesterol—guide clinical decisions according to local capacity. Universal lipid screening is not a WHO priority unless the health system has sufficient capacity to ensure that testing leads reliably to effective treatment. This perspective helps explain why global lipid-testing patterns are uneven: WHO does not reject cholesterol measurement; it prioritises deploying it where the result can be interpreted, acted upon, and linked to affordable treatment.

7. Limitations and Blind Spots

The standard lipid panel is an indispensable tool, but it is not a flawless one. An appraisal of its limitations is essential to a balanced evaluation. These limitations fall into three categories: those intrinsic to the panel itself, problems created by misuse or misinterpretation, and broader system failures that prevent lipid testing from translating into effective prevention.

7.1 Limitations of the Panel Itself

LDL-C and apolipoprotein B discordance. LDL-C measures the cholesterol mass carried within low-density lipoprotein particles, not the number of circulating atherogenic particles. Apolipoprotein B (apoB) provides a closer estimate of the number of major atherogenic apoB-containing particles, because LDL, VLDL, IDL remnants, and Lipoprotein(a) [Lp(a)] each generally carry one apoB-100 molecule. This distinction matters: in people with insulin resistance, type 2 diabetes, obesity, metabolic syndrome, or hypertriglyceridaemia, LDL particles may be smaller, more numerous, or relatively cholesterol-depleted, so LDL-C can underestimate the true atherogenic particle burden. A clinically meaningful proportion of patients show discordance between LDL-C and apoB, particularly in cardiometabolic risk states. Across multiple cohort and discordance analyses, apoB often performs better than LDL-C as a marker of ASCVD risk when the two measures disagree. Major consensus statements and guidelines increasingly recognise apoB as an alternative or secondary treatment target, especially in people with diabetes, obesity, high triglycerides, metabolic syndrome, or residual risk despite apparently controlled LDL-C. Because apoB is not included in most standard lipid panels, this discordance remains an important and often invisible source of risk misclassification.

Non-HDL-C and remnant cholesterol. Non-HDL-C, calculated as total cholesterol minus HDL-C, captures the cholesterol content of the major apoB-containing atherogenic lipoproteins, including LDL, IDL, VLDL remnants, and Lp(a). It requires no additional blood test and is increasingly reported alongside the standard panel. Non-HDL-C performs at least as well as LDL-C for risk prediction in many populations and may perform better when triglycerides are elevated, because it includes cholesterol carried in triglyceride-rich lipoproteins and their remnants. ESC/EAS, ACC/AHA, and other guidelines recognise non-HDL-C as a complementary marker or secondary target, particularly in people with hypertriglyceridaemia, diabetes, obesity, or metabolic syndrome. Despite this, non-HDL-C is not universally reported, highlighted, or acted upon; in many clinical encounters, LDL-C remains the dominant number discussed with patients, even when non-HDL-C or apoB would provide more useful context. Remnant cholesterol, a subset of non-HDL-C, represents cholesterol carried in triglyceride-rich lipoproteins and their partially lipolysed remnants. Observational and genetic studies support a role for remnant cholesterol in ASCVD risk, particularly in people with elevated triglycerides. However, the evidence base for routine

clinical measurement and treatment targeting of remnant cholesterol is less mature than for LDL-C, non-HDL-C, or apoB. For now, remnant cholesterol is best understood as an emerging marker of residual risk rather than a routine component of standard lipid assessment.

Lipoprotein(a) exclusion. Lipoprotein(a) [Lp(a)] is a genetically determined risk factor for ASCVD and calcific aortic valve stenosis, with evidence also linking elevated Lp(a) to ischaemic stroke risk. It is not captured by the standard lipid panel, is largely refractory to lifestyle modification, and is not meaningfully lowered by statins, though PCSK9 inhibitors produce modest reductions. Approximately one in five people globally have elevated Lp(a). Levels around 50 mg/dL or 125 nmol/L are associated with increased long-term ASCVD risk, while substantially higher levels, such as around 250 nmol/L, are associated with at least a two-fold increase in long-term heart attack or stroke risk. The 2026 ACC/AHA/multisociety dyslipidaemia guideline recommends that Lp(a) be measured at least once in adulthood. Yet in most health systems, Lp(a) testing remains outside the routine panel. This makes it one of the largest single-marker gaps in standard lipid assessment. A person may therefore have a lipid panel that appears reassuring while carrying a substantial inherited risk that remains invisible unless Lp(a) is specifically ordered.

Insulin resistance and metabolic syndrome. The characteristic lipid pattern of insulin resistance and metabolic syndrome includes elevated triglycerides, low HDL-C, increased triglyceride-rich lipoproteins, small dense LDL particles, and often a normal or only modestly elevated LDL-C. The standard panel may therefore appear less concerning than the person's actual cardiometabolic risk warrants. The panel does not directly measure insulin resistance, hyperinsulinaemia, glucose intolerance, HbA1c, visceral adiposity, waist circumference, or hepatic fat. It may provide clues, especially through triglycerides, HDL-C, and non-HDL-C, but it cannot diagnose the underlying metabolic state.

Inflammation and residual risk beyond lipids. Even with effective LDL-C lowering, residual cardiovascular risk persists. Some of that residual risk is related to non-lipid pathways, including inflammation, thrombosis, blood pressure, glycaemia, kidney disease, smoking, and established plaque burden. The CANTOS trial demonstrated that targeting the interleukin-1 β inflammatory pathway with canakinumab reduced cardiovascular events without lowering lipid levels, supporting the concept that inflammatory risk can be biologically and clinically relevant independent of LDL-C. High-sensitivity C-reactive protein (hs-CRP) is a widely studied inflammatory biomarker associated with cardiovascular events. However, hs-CRP is not part of the standard lipid panel and is not recommended as a universal screening test for all adults. Its role is more selective: it may help refine risk in certain intermediate-risk patients or identify residual inflammatory risk in selected clinical contexts. The key point is not that hs-CRP should replace lipid testing, but that the lipid panel does not capture all biologically relevant cardiovascular risk pathways.

Coronary artery calcium and subclinical atherosclerosis. Coronary artery calcium (CAC) scoring is not a replacement for lipid testing, but it illustrates an important limitation of blood-based risk markers. Lipid values estimate exposure and risk; CAC detects established calcified plaque. A person may have apparently acceptable lipid values but substantial

subclinical atherosclerosis, especially after years of modest but cumulative exposure to multiple risk factors. Conversely, a CAC score of zero generally indicates low short-term event risk, but it does not eliminate lifetime risk, particularly in younger individuals whose plaque may be non-calcified or in people with major inherited risk factors. CAC is therefore useful in selected patients when risk remains uncertain after standard clinical and lipid assessment. Its role highlights a broader principle: lipid panels estimate risk, but they do not directly image the arterial consequences of that risk.

"Normal" ranges versus individualised targets. Laboratory reference intervals are often derived from population distributions or broad interpretive categories. They are not the same as personalised treatment thresholds. An LDL-C value that appears "normal" on a laboratory report may still exceed the appropriate target for a person with established ASCVD, diabetes, chronic kidney disease, familial hypercholesterolaemia, high Lp(a), or multiple risk factors. The word "normal" can therefore create false reassurance when lipid results are not interpreted in relation to absolute risk, cumulative exposure, treatment goals, and clinical context.

7.2 Misuse, Over-Interpretation, and Misunderstanding

Over-testing. In high-income health systems, repeat lipid panels are frequently performed in stable, low-risk individuals without clear evidence that the result will change management. The Oxfordshire audit documented a more than 15-fold rise in lipid testing over two decades, with repeat monitoring accounting for much of this growth and a substantial proportion of tests judged potentially unnecessary. Frequent retesting can consume resources, generate borderline or biologically variable results, and prompt unnecessary anxiety or treatment adjustments. This is not a failure of the lipid panel itself. It is a failure to use the test with a clear clinical question. A baseline lipid assessment is valuable. Repeating the same assessment habitually in a stable low-risk person may not be.

Under-testing. At the same time, high-risk groups remain systematically under-screened. These include young adults with familial hypercholesterolaemia, many of whom remain undiagnosed; people with a strong family history of premature ASCVD; socioeconomically disadvantaged populations; rural communities; uninsured or underinsured groups; and some high-risk ethnic groups in whom ASCVD develops earlier or at lower conventional LDL-C thresholds. Globally, under-testing is even more pronounced. In a multi-country analysis of low- and middle-income countries, among adults meeting WHO PEN criteria for diagnostic cholesterol testing, only 39.7% had ever had their cholesterol measured. This figure illustrates how far "routine" cholesterol testing is from being universal. The same lipid panel can therefore be overused in some low-risk populations and underused in populations where the potential benefit is greatest.

Misunderstanding of "normal" results. Many patients, and some non-specialist clinicians, interpret a lipid panel within the laboratory reference range as indicating the absence of cardiovascular risk. This misunderstanding is common when results are communicated as isolated numbers rather than as part of absolute cardiovascular risk. A lipid panel can appear

"normal" while a person has uncontrolled hypertension, active smoking exposure, diabetes, chronic kidney disease, high Lp(a), strong family history, or established subclinical plaque. Calling the panel a "standard" or "routine" cholesterol test may unintentionally reinforce the perception that it is a complete cardiovascular risk assessment rather than one part of a broader evaluation. Survey evidence suggests that many adults cannot accurately recall their cholesterol values, and a substantial minority of people with elevated LDL-C are unaware of their status. This gap between testing and understanding is directly relevant to the terminology discussion.

7.3 Broader System Failures

Even a well-performed and appropriately interpreted lipid panel may fail to improve outcomes if the surrounding clinical system does not support timely action. Therapeutic inertia is common: lipid-lowering therapy may not be intensified even when LDL-C, non-HDL-C, or apoB remain above recommended thresholds. Real-world registries consistently show that a sizable proportion of patients with established ASCVD remain above recommended LDL-C targets, even when statins are prescribed. Abnormal results may be recorded but not explained. Patients may receive a printout without a risk calculation, treatment discussion, or follow-up plan. Conversely, some health systems reward testing volume more easily than longitudinal prevention, leading to repeated measurement without meaningful intervention. Other system failures include poor integration of lipid results with blood pressure, glycaemia, kidney function, smoking status, family history, and medication access; lack of structured recall systems; fragmented care between primary care and specialists; and inequitable access to follow-up testing and therapy. In these cases, the limitation is not the panel itself. The limitation is the pathway around it.

7.4 Implications for Baseline Framing

These limitations do not diminish the substantial value of the standard lipid panel. They clarify its role. It is valuable because it provides a low-cost, reproducible, scalable first layer of cardiovascular risk information. Its limitations arise when that first layer is treated as the whole picture. Reframing the test as a baseline lipid assessment or foundational lipid evaluation may help preserve its value while reducing false reassurance. Such language communicates that the panel is essential, but not final; actionable, but context-dependent; proven, but not exhaustive. A baseline framing may also help distinguish appropriate from inappropriate use. It supports early measurement, risk calculation, and treatment initiation when indicated. It discourages habitual repeat testing without a clinical question. It reminds clinicians and patients that further context may be needed: blood pressure, glycaemic status, smoking exposure, kidney function, family history, apoB, Lp(a), inflammation markers, or imaging in selected cases. The appropriate conclusion is therefore not that standard cholesterol testing has failed, but that it should be understood and communicated more precisely. It remains the necessary first layer of cardiovascular risk evaluation, upon which further testing, risk context, and clinical judgment can be built.

8. Baseline vs Standard Terminology: Critical Evaluation

The language used to describe cholesterol testing shapes clinical expectations, patient understanding, and policy design. Terms such as "standard", "routine", "baseline", "foundational", and "initial" are not neutral. Each carries assumptions about what the lipid panel is, what it can tell us, and what should happen after the result is known. This section compares the major terms used to describe cholesterol testing and evaluates which framing best serves modern cardiovascular prevention.

8.1 What Do Current Terms Signify?

Each term in common use carries distinct connotations that influence how clinicians, patients, laboratories, and health systems understand the role of the lipid panel.

Standard lipid panel. "Standard lipid panel" is useful laboratory shorthand. It describes the conventional panel used in routine clinical practice: total cholesterol, HDL-C, triglycerides, calculated LDL-C, and often calculated non-HDL-C. Its strength is familiarity. Clinicians, laboratories, payers, and health systems understand what it usually means. Its weakness is interpretive. In public-facing use, "standard" may be heard as "complete" or "sufficient". A patient told that their "standard cholesterol test" is normal may reasonably assume that their lipid-related cardiovascular risk has been fully assessed. That assumption is not always correct, particularly when apoB discordance, elevated Lipoprotein(a), strong family history, metabolic syndrome, hypertension, smoking, diabetes, chronic kidney disease, or subclinical plaque is present.

Routine cholesterol test. "Routine cholesterol test" connotes a test performed regularly or as part of ordinary care. It is operationally useful in screening programmes, primary-care workflows, and public-facing health invitations. Its weakness is that "routine" can imply habit rather than purpose. If the reason for testing is not made explicit, the term may encourage repeated measurement without a clear clinical question. It can also make the test sound mundane, even though abnormal results may carry major implications for prevention, treatment, and family risk.

Baseline lipid assessment. "Baseline lipid assessment" implies a starting point. It positions the lipid panel as an initial measurement against which risk, treatment response, and future change can be interpreted. This framing has several advantages. It avoids implying completeness. It communicates that the test is important but not final. It also fits clinical reality: untreated lipid values are used to estimate risk, identify dyslipidaemia, guide therapy initiation, assess percentage LDL-C reduction, and monitor response over time. The potential pitfall is that "baseline" may be misread as "one-and-done"; this can be avoided by explaining that repeat testing depends on age, risk status, treatment changes, and clinical context. For

example, in low-risk adults a repeat lipid panel every five years may be reasonable, while more frequent testing is appropriate when therapy is initiated or adjusted, metabolic health shifts, or new risk factors emerge.

Foundational lipid assessment. “Foundational lipid assessment” suggests that the panel provides essential but incomplete information. It captures the lipid panel’s enabling role in cardiovascular prevention without overstating its scope. This term is especially useful in professional reports, policy discussion, and educational materials. Its weakness is familiarity: “foundational” is accurate but less common in clinical practice and may require explanation.

Initial lipid-risk assessment. “Initial lipid-risk assessment” is the most precise term in formal clinical language. It explicitly describes the test as an assessment of the lipid component of cardiovascular risk, performed early in the prevention pathway, and interpreted alongside other risk information. Its weakness is practicality. It is accurate but too long for most public-facing communication. It works best in guidelines, professional education, and structured clinical pathways.

Baseline cardiovascular risk check. “Baseline cardiovascular risk check” may be the most useful patient-facing phrase. It frames cholesterol testing as part of a broader heart-risk conversation rather than as a standalone verdict. Its strength is clarity: it implies that cholesterol, blood pressure, diabetes status, smoking exposure, family history, and other risk factors belong together. Its weakness is that it must be defined carefully: a baseline cardiovascular risk check should not be reduced to cholesterol alone.

8.2 Evidence and Guideline Usage

Major guidelines and professional documents already use language that is compatible with a baseline or initial-assessment framing, even when they do not explicitly reject the term “standard.”

The ESC/EAS dyslipidaemia guidelines define “baseline” LDL-C as the LDL-C level in a person not taking LDL-C-lowering medication. This is a functional and clinically useful concept: it separates untreated lipid exposure from treated lipid status and provides a reference point for monitoring response. NICE NG238 uses the phrase “initial lipid measurement” when describing lipid testing before formal cardiovascular risk calculation. This language already frames the lipid panel as an entry point into risk estimation rather than a complete assessment in isolation.

ACC/AHA guidance also supports a layered model of lipid evaluation. Contemporary guidance emphasises lifetime lipid measurement, LDL-C thresholds, treatment response, selective use of coronary artery calcium when risk remains uncertain, Lipoprotein(a) measurement at least once in adulthood, and apoB testing in selected people with residual or cardiometabolic risk. This is not a single-test model. It is a structured, risk-based approach in which the standard lipid panel remains central but not exhaustive.

The term “baseline” aligns with several evidence-based observations. A lipid panel obtained in young adulthood can help stratify long-term risk and guide future testing frequency. Untreated LDL-C values help identify severe hypercholesterolaemia, familial hypercholesterolaemia, and treatment eligibility. Baseline LDL-C is necessary for assessing percentage response to lipid-lowering therapy. A baseline lipid result may remain informative for several years in low-risk individuals, reducing the need for unnecessary repeat testing. Additional testing, such as apoB, Lipoprotein(a), hs-CRP, or coronary artery calcium, may be appropriate when standard lipid values do not fully explain risk. For these reasons, “baseline” more accurately reflects how lipid testing functions in modern prevention: it provides the first structured lipid-risk layer, not a final cardiovascular verdict.

8.3 Which Terms Fit Which Contexts?

The optimal terminology depends on audience and purpose.

Clinical settings. For clinicians, “baseline lipid profile” or “foundational lipid assessment” places the test within a broader risk-evaluation framework. It preserves the test’s importance while avoiding the implication that the panel alone is sufficient. This framing also aligns with clinical workflow. A lipid panel is used to establish untreated risk, support risk calculation, identify dyslipidaemia, guide therapy, and monitor response. In selected patients, it may prompt additional evaluation: apoB when particle burden is uncertain, Lipoprotein(a) when inherited risk is relevant, or CAC when treatment decisions remain unclear.

Patient-facing communication. For patients, “baseline” is intuitive. It suggests a starting measurement rather than a final answer. A phrase such as “baseline cardiovascular risk check” may be preferable to “standard cholesterol test” because it makes clear that cholesterol testing belongs inside a broader assessment. It also creates room for clearer result communication: “Your LDL-C is one part of your risk picture. We also need to consider your blood pressure, smoking exposure, blood sugar, family history, kidney function, and whether any additional lipid markers are relevant”. This is more useful than simply saying “Your cholesterol is normal”. The first statement supports understanding and action. The second may create false reassurance.

Public-health and policy settings. For public-health programmes, “routine lipid panel” can still be useful as an operational descriptor. It tells programme designers, laboratories, and payers what test is being delivered. However, when communicating results or designing prevention pathways, “baseline lipid screening” or “initial lipid-risk assessment” is more accurate. These terms signal that the test is the beginning of a risk-assessment process, not the whole process.

Low-resource settings. The terminology question is especially sensitive in low-resource settings. In high-resource systems, calling the lipid panel “baseline” can help reduce false reassurance and support layered risk assessment. In settings where apoB, Lipoprotein(a), hs-CRP, or coronary artery calcium are not feasible, however, overemphasising incompleteness may create unrealistic expectations without offering actionable alternatives.

In these settings, the priority may be different: making baseline lipid testing available, affordable, interpretable, and linked to treatment. The standard lipid panel should not be framed as inadequate where even one lipid measurement remains inaccessible to many people. A globally responsible terminology strategy should therefore distinguish between two messages: in high-resource settings, **the standard lipid panel is essential, but it may not be the full risk picture**; in low-resource settings, **a baseline lipid measurement is valuable and should be integrated into total-risk assessment where feasible**. Both statements can be true.

The table below consolidates the terminology options discussed above as a quick reference.

Terminology	Best Fit	Potential Pitfall
Standard lipid panel	Laboratory catalogues, conventional clinical shorthand, payer coding	May be heard by patients as complete or sufficient
Routine cholesterol test	Screening invitations, operational workflows, age-based preventive programmes	May encourage habitual repetition unless the purpose is clear
Baseline lipid assessment	First adult measurement, treatment initiation, longitudinal comparison, low-risk screening	May be mistaken as "one-and-done" unless repeat criteria are explained
Foundational lipid assessment	Professional reports, policy framing, clinician education	Less familiar term; may require explanation
Initial lipid-risk assessment	Guidelines, preventive cardiology pathways, formal clinical evaluation	Accurate but too long for most public messaging
Baseline cardiovascular risk check	Patient-facing communication and public-health education	Must clarify that lipids are only one part of the check

8.4 Does Terminology Change Practice?

There is limited direct evidence that relabelling cholesterol testing would change clinician behaviour, patient understanding, testing frequency, or cardiovascular outcomes. No randomised trial has compared terms such as "standard cholesterol test", "routine cholesterol test", "baseline lipid assessment", or "foundational lipid evaluation".

However, evidence from adjacent areas of risk communication suggests that labels and framing can influence how people interpret results and decide whether action is needed. Communication that presents numerical results in context, explains absolute risk, and links the result to a concrete next step is generally more useful than categorical labels such as "normal", "borderline", or "high" in isolation. This matters because lipid panels are often interpreted through categories. A patient may see "normal" and assume safety. A clinician may see a modest elevation and repeat the test without calculating absolute risk. A health system may reward measurement without ensuring follow-through.

Terminology alone will not fix these problems. But terminology can support better expectations. A "baseline lipid assessment" framing may help communicate that the test is important, the result should be interpreted in context, repeat testing should have a purpose, additional risk information may be needed in selected patients, and a normal result is reassuring only within the broader risk picture. The evidence for this remains indirect. Direct research is needed to test whether lipid-test terminology affects patient understanding, clinician ordering behaviour, appropriate statin use, repeat testing frequency, or follow-up completion. This should be considered an unanswered research question, not a settled behavioural intervention.

8.5 Conclusion on Terminology

The evidence does not support abandoning the term **standard cholesterol test** in all contexts. It remains useful as laboratory shorthand and as a practical description of the conventional lipid panel.

However, the evidence and guideline landscape do support a more deliberate communication shift. In clinical and public-health contexts, the lipid panel is better framed as a baseline or foundational assessment: essential, proven, and actionable, but not a complete cardiovascular risk evaluation.

The most appropriate language is context-specific. In clinical settings, **baseline lipid profile** or **foundational lipid assessment** accurately positions the panel within broader risk evaluation. In patient-facing materials, **baseline cardiovascular risk check** may better communicate that cholesterol testing is one part of heart-risk assessment. In policy documents, **initial lipid-risk assessment** captures the conditional, starting-point nature of the test. In operational laboratory settings, **standard lipid panel** remains clear and useful.

What matters most is not replacing one phrase with another everywhere. What matters is the expectation created by the phrase. Patients, clinicians, and policy makers should understand the standard lipid panel as an essential first layer of cardiovascular risk assessment, not the complete picture of risk. It is not obsolete. It is not sufficient in every context. It is a proven baseline from which modern prevention can proceed.

9. From Baseline to Enhanced: What Modern Lipid-Risk Assessment Should Include

A contemporary lipid-risk assessment must reconcile two realities. The first is that the basic lipid panel, total cholesterol, HDL-C, triglycerides, calculated LDL-C where valid, and calculated non-HDL-C, remains an indispensable, low-cost, and globally scalable foundation. The second is that modern evidence has identified important sources of cardiovascular risk that this panel does not capture. A layered, context-dependent approach is therefore more accurate than a single uniform test for all people in all settings.

The appropriate question is not whether the standard lipid panel should be replaced. It should not. The question is **how the baseline panel should be embedded within a broader risk assessment pathway that reflects clinical need, system capacity, and the consequences of missing risk in selected patients.**

9.1 A Tiered Framework for Lipid-Risk Assessment

Health systems differ dramatically in resources, laboratory capacity, disease burden, workforce availability, and treatment access. A tiered model allows the intensity of assessment to be matched to clinical need and system capability while preserving the basic lipid panel as the universal starting point where testing capacity exists.

Tier 1: Basic Baseline Lipid Panel

The basic baseline panel should include total cholesterol, HDL-C, triglycerides, LDL-C (calculated where valid or measured directly where required), and non-HDL-C (calculated and reported routinely). This panel can usually be performed on a non-fasting sample, is inexpensive relative to advanced testing, and is widely automated in health systems with functioning clinical chemistry infrastructure. It provides the lipid inputs for most laboratory-based cardiovascular risk calculators and supports diagnosis, treatment initiation, and monitoring.

In health systems with basic laboratory capacity, this panel should be available to adults as part of initial cardiovascular risk assessment. Where universal access is not yet feasible, testing should be prioritised for higher-risk groups: people with hypertension, diabetes, chronic kidney disease, obesity, smoking exposure, strong family history, suspected familial hypercholesterolaemia, established cardiovascular disease, or other clinical risk indicators.

Repeat testing should be determined by risk status and clinical purpose. In low-risk individuals with reassuring baseline results, repeat testing can often occur at multi-year intervals. Testing should be more frequent when lipid-lowering therapy is initiated or adjusted, when metabolic status changes, when diabetes or kidney disease develops, or when a new cardiovascular diagnosis is made.

Tier 2: Enhanced Lipid Assessment

Enhanced lipid assessment is not required for everyone at every encounter. It is most useful in higher-resource settings or selected patients whose standard lipid panel may underestimate risk.

Apolipoprotein B. Apolipoprotein B (apoB) provides a closer estimate of the number of major atherogenic apoB-containing particles because LDL, VLDL, IDL remnants, and Lipoprotein(a) [Lp(a)] each generally carry one apoB-100 molecule. ApoB measurement is especially useful when LDL-C may not accurately reflect particle burden. This includes people with elevated triglycerides, diabetes, metabolic syndrome, obesity, chronic kidney disease, very low achieved LDL-C, suspected discordance between LDL-C and clinical risk, or residual risk despite apparently acceptable LDL-C and non-HDL-C. Guideline consensus and discordance analyses support apoB as a more accurate marker than LDL-C in selected cardiometabolic-risk populations. The practical barrier is rarely analytical feasibility; apoB immunoassays are automated and standardised in many laboratory systems. The larger barriers are ordering habits, reimbursement, clinician familiarity, and integration into treatment targets.

Lipoprotein(a). Lp(a) is a genetically determined, largely stable, and independent risk factor for ASCVD and calcific aortic valve stenosis. It is not captured by the standard lipid panel and is not meaningfully lowered by statins. In high-resource settings, once-in-adulthood Lp(a) measurement is increasingly guideline-supported. The 2026 ACC/AHA/multisociety dyslipidaemia guideline recommends Lp(a) measurement at least once in adulthood. In lower-resource settings, targeted Lp(a) testing may be prioritised for people with premature ASCVD, familial hypercholesterolaemia, a strong family history of premature ASCVD, unexplained aortic stenosis, or risk that appears disproportionate to standard lipid values.

Levels around 50 mg/dL or 125 nmol/L are associated with increased long-term ASCVD risk. Very high Lp(a), particularly levels around or above 180 mg/dL or 430 nmol/L, may confer lifetime ASCVD risk comparable to heterozygous familial hypercholesterolaemia. Because Lp(a) is usually measured once rather than repeatedly, the cost of a single test is amortised over decades. Its main value lies in identifying people who may warrant earlier and more intensive global risk-factor control, particularly sustained LDL-C and apoB lowering, blood-pressure control, family cascade awareness, and appropriate follow-up. Direct outcome-trial evidence for therapies that specifically lower Lp(a) remains an evolving area. However, the genetic, observational, and mechanistic evidence supporting Lp(a) as a risk marker is strong enough that many contemporary guidelines now support at least one measurement in adulthood where feasible.

Tier 3: Broader Risk Context Package

A lipid panel interpreted without broader clinical context cannot generate a valid absolute cardiovascular risk estimate. The broader risk context package should accompany every baseline lipid assessment wherever feasible. It should include blood pressure, glycaemic

status (HbA1c or fasting glucose), smoking and nicotine exposure, family history of premature ASCVD, kidney function (eGFR and albuminuria where feasible), BMI and waist circumference, current medication use, and history of ASCVD, diabetes, chronic kidney disease, inflammatory disease, premature menopause, pregnancy complications, or other risk-enhancing conditions.

These factors are not optional extras. They determine absolute risk, shape the interpretation of lipid values, and often determine whether treatment is indicated. A lipid panel alone can identify severe dyslipidaemia, but most prevention decisions require the integration of lipids with these broader risk factors. In low-resource settings, this broader package may be more feasible than advanced lipid testing or imaging. Blood pressure measurement, diabetes screening, smoking assessment, and family-history capture can substantially improve risk stratification even when apoB, Lp(a), or CAC scoring are unavailable.

Tier 4: Advanced Risk Stratification

Advanced risk stratification is appropriate for selected patients in higher-resource settings, particularly when treatment decisions remain uncertain after standard clinical and lipid assessment.

Coronary artery calcium. Coronary artery calcium (CAC) scoring detects established calcified coronary plaque. It is not a replacement for lipid testing and should not be used as a population-wide screening tool. Its role is selective. CAC scoring may be considered when the treatment decision remains uncertain after clinician-patient discussion, particularly in borderline or intermediate estimated risk. A CAC score of zero can support deferring statin therapy in selected low- or borderline-risk adults, while a clearly positive CAC score can reclassify risk upward and support more intensive prevention. CAC is especially useful when risk calculators and lipid results do not align with clinical suspicion. CAC has limitations: it detects calcified plaque, not early non-calcified plaque; it is less informative in very young adults; and it should not be used to dismiss major inherited risk factors or severe hypercholesterolaemia. It also requires CT infrastructure, trained interpretation, radiation exposure, and reimbursement capacity, making it unsuitable as a universal global tool.

High-sensitivity C-reactive protein. High-sensitivity C-reactive protein (hs-CRP) may be considered in selected patients to refine risk estimates or identify residual inflammatory risk. Evidence from JUPITER and CANTOS supports the biological and clinical relevance of inflammatory pathways in ASCVD. However, hs-CRP is not a universal screening test and should not be used to substitute for lipid measurement. Its most appropriate use is selective: intermediate-risk patients, patients with inflammatory disorders, or patients with residual risk despite controlled LDL-C where the result would plausibly influence management.

9.2 Baseline vs Enhanced Testing Framework

Assessment Layer	Components	When	Resource Level	Evidence Strength
Foundation: baseline lipid panel	TC, HDL-C, TG, calculated LDL-C where valid, calculated non-HDL-C	Initial adult lipid-risk assessment; repeat according to risk, treatment changes, and prior results	Universal goal; may require prioritisation in low-resource systems	Cohort evidence; randomised trial evidence for lipid-lowering decisions enabled by testing; guideline consensus
Enhanced lipid assessment: apoB	ApoB	Diabetes, metabolic syndrome, obesity, elevated TG, low achieved LDL-C, CKD, suspected LDL-C/particle discordance, residual risk	Medium to high	Discordance analyses; observational data; guideline and consensus support
Enhanced lipid assessment: Lp(a)	Lp(a), preferably in nmol/L	Once in adulthood where feasible; targeted testing where resources are limited	Medium to high	Mendelian randomisation; cohort studies; guideline consensus; direct outcome-trial evidence for Lp(a)-specific drugs pending
Risk context package	BP, HbA1c or glucose, smoking/nicotine exposure, family history, eGFR, albuminuria, BMI/waist	With every baseline lipid assessment	Universal	Guideline consensus; cohort studies; INTERHEART and global risk-factor evidence
Advanced risk stratification: imaging	CAC score	Borderline or intermediate estimated risk when treatment decision remains uncertain	High	Cohort studies; risk-reclassification evidence; guideline support
Inflammatory context	hs-CRP	Selected intermediate-risk or residual-risk patients	High	JUPITER, CANTOS, cohort evidence, selective guideline use

9.3 Implementation Considerations

The basic baseline panel is feasible in health systems with functioning clinical chemistry infrastructure, reliable reagents, calibrated analysers, quality-control systems, trained personnel, and mechanisms for reporting results back to clinicians and patients. It is not universally available in all primary-care settings globally. Where laboratory capacity is constrained, prioritisation of higher-risk groups is more realistic than universal adult testing.

Adding apoB to lipid assessment is analytically straightforward in many higher-resource systems. ApoB immunoassays are standardised, automated, and comparable in operational complexity to other routine immunoassays. The main barriers are institutional rather than

technical: clinician familiarity, ordering habits, laboratory menus, reimbursement, and uncertainty about how to act on discordant results. In populations with high prevalence of diabetes, metabolic syndrome, obesity, or hypertriglyceridaemia, broader apoB use may be clinically valuable, but formal health-economic modelling across diverse settings remains incomplete.

Universal once-in-adulthood Lp(a) testing faces a different implementation challenge. Assay standardisation has improved, but variation persists, and reporting in nmol/L versus mg/dL continues to cause confusion. Reimbursement is inconsistent. Clinicians may also be uncertain about how to manage a high Lp(a) result in the absence of approved Lp(a)-specific outcome-proven therapy. Nevertheless, because the test is usually performed once, the per-person lifetime cost may be modest in high-resource systems. The immediate clinical value lies in risk reclassification, earlier LDL-C and apoB lowering, family cascade awareness, and identification of patients who may qualify for future targeted therapies or clinical trials.

CAC scoring is more expensive, less scalable, and more infrastructure-dependent than blood-based assessment. Its appropriate use is selective. It is most valuable when risk is uncertain and the result would change the treatment decision. CAC should not be used as a substitute for lipid testing, nor should a zero score be interpreted as lifetime immunity from ASCVD.

The risk of overtreatment must be acknowledged. Expanding testing to include apoB or Lp(a) may reclassify some individuals into higher-risk categories and may increase statin or combination-therapy use. However, the clinical hazard of over-identifying risk is generally smaller than the hazard of under-identifying it, particularly when enhanced testing is applied to populations with higher baseline risk rather than unselected low-risk groups. Shared decision-making, clear action thresholds, and evidence-based treatment algorithms can mitigate inappropriate treatment escalation. The greater clinical concern in many settings remains the under-identification of high-risk individuals, especially those with familial hypercholesterolaemia, high Lp(a), diabetes, chronic kidney disease, or strong family history. Implementation should therefore pair expanded assessment with clear criteria for when testing adds value, and with systems that ensure abnormal results lead to appropriate prevention.

9.4 Global Minimum

In low-resource settings, the priority is not universal advanced lipid assessment. The priority is to make baseline cardiovascular risk assessment available, affordable, interpretable, and linked to effective treatment. A pragmatic global minimum includes blood pressure measurement, smoking and nicotine exposure assessment, diabetes screening where feasible, basic lipid measurement where laboratory capacity exists, family history of premature cardiovascular disease, access to affordable generic statins for eligible patients, total-risk assessment using WHO PEN or HEARTS tools, and non-laboratory risk charts when cholesterol testing is unavailable.

Where lipid testing is feasible, total cholesterol and HDL-C can support risk calculation, while triglycerides and calculated non-HDL-C add useful context. Where full lipid panels are not feasible, total-risk tools that do not require laboratory measurement may still support treatment decisions, especially for blood pressure control, smoking cessation, and statin use in clearly high-risk individuals. The term "baseline" in these settings should communicate that a first lipid measurement is valuable and actionable, not that it is inadequate. In many health systems, the most important policy goal is still to ensure that at least basic lipid measurement is available to people who meet evidence-based testing criteria and that results are linked to affordable prevention.

9.5 Practical Conclusion

This framework preserves the central value of standard cholesterol testing while making its role more precise. The baseline lipid panel remains the first indispensable layer. Modern prevention requires knowing when that layer is enough, when it should be repeated, and when it should be expanded.

10. Balanced Conclusion

10.1 Synthesis of Evidence

The evidence assembled in this report supports a balanced conclusion: **routine cholesterol testing remains one of the most important achievements in cardiovascular prevention, but its role is best understood today as foundational rather than complete. The standard lipid panel should be preserved, used more consistently where underused, interpreted more carefully where over-relied upon, and expanded selectively when clinical context warrants.**

Routine lipid panels are valuable and historically important. They enabled the epidemiological discovery of the cholesterol-ASCVD association, supported the development of cardiovascular risk prediction, were instrumental in the deployment and monitoring of statin therapy, and remain central to cardiovascular prevention worldwide. From Framingham and the Seven Countries Study to the Lipid Research Clinics trial, the statin era, and the Cholesterol Treatment Trialists' meta-analyses, lipid measurement has helped turn cardiovascular prevention from broad lifestyle advice into measurable, treatable, and monitorable risk reduction.

EVIDENCE STRENGTH: historical analysis; prospective cohort evidence; randomised trial evidence for lipid-lowering decisions enabled by testing; guideline consensus.

They remain foundational tools. No equally scalable laboratory alternative has replaced the basic lipid panel for population-level lipid screening or initial lipid-risk stratification. Total cholesterol, HDL-C, triglycerides, calculated LDL-C where valid, and calculated non-HDL-C provide essential information that cannot be obtained from history or examination alone. The lipid panel is inexpensive, widely automated, reproducible, clinically familiar, and actionable.

EVIDENCE STRENGTH: guideline consensus; cohort evidence; health-economic modelling; randomised trial evidence for lipid-lowering treatment strategies guided by lipid measurement.

They may not always be sufficient for individualised modern prevention. The standard lipid panel does not capture all relevant cardiovascular risk. Discordance between LDL-C and apoB, inherited Lipoprotein(a)-mediated risk, metabolic syndrome, chronic kidney disease, subclinical atherosclerosis, and residual inflammatory risk mean that a "normal" lipid panel may be incomplete in selected patients. This does not mean the panel is obsolete; it means its result should be interpreted in context.

EVIDENCE STRENGTH: discordance analyses; cohort studies; Mendelian randomisation; guideline consensus.

Baseline or foundational framing is more accurate than complete-test framing. The evidence does not yet prove that changing terminology improves clinical outcomes, but the language of "baseline", "foundational", or "initial lipid-risk assessment" better reflects how the panel functions in modern prevention. Calling it a "standard cholesterol test" remains useful as laboratory shorthand, but in public-facing or clinical communication it may imply sufficiency. Baseline framing preserves the test's value while reducing the risk of false reassurance.

EVIDENCE STRENGTH: guideline language; clinical reasoning; risk-communication theory; expert opinion; direct implementation evidence lacking.

Modern prevention requires layered risk assessment in selected populations. The basic lipid panel should be interpreted alongside blood pressure, glycaemic status, smoking or nicotine exposure, family history, kidney function, body composition, and medication history. ApoB, Lipoprotein(a), coronary artery calcium scoring, and hs-CRP should be considered selectively when feasible and likely to change management. This layered approach does not mean testing everything in everyone; it means beginning with the baseline panel, then escalating assessment based on risk, uncertainty, and health-system capacity.

EVIDENCE STRENGTH: guideline recommendations; cohort studies; Mendelian randomisation for Lipoprotein(a); randomised trial evidence for LDL-lowering and selected inflammatory-risk interventions.

Taken together, these conclusions do not advocate abandoning or diminishing routine cholesterol testing. On the contrary, the achievements of lipid testing represent one of the most successful translational stories in modern medicine. The more precise conclusion is that the same test that was revolutionary in 1972, when the Friedewald equation first made LDL-C estimation widely available, is today best understood as the essential but incomplete first chapter of a modern cardiovascular risk story.

The language used to describe the test matters. Calling a lipid panel a "baseline assessment" rather than a "standard test" is not merely semantic. It shapes expectations, guides follow-up, and may reduce both unnecessary repeat testing and the false reassurance that can arise from a "normal" result. It preserves public trust by being honest about what the test can and cannot tell us.

A critical proviso is necessary. The greatest addressable source of preventable cardiovascular risk is not the incompleteness of the standard panel, but the failure to apply validated risk assessment systematically to those who most need it. This is especially true in low- and middle-income countries, in underserved communities, in high-risk ethnic groups, and among people with undetected lipoprotein(a), familial hypercholesterolaemia, or other inherited risk. In settings where even a single lipid measurement remains out of reach, the priority is to make baseline testing available, affordable, interpretable, and linked to treatment, not to overemphasise what the panel cannot do.

10.2 Implications for Clinicians and Health Systems

Several practice implications follow from this analysis.

First, clinicians should resist treating a lipid panel within the laboratory reference range as a reassuring endpoint. Lipid values must be interpreted within the patient's total cardiovascular risk profile. Validated risk calculators should be applied systematically rather than opportunistically, and lipid results should be considered alongside blood pressure, glycaemic status, smoking exposure, kidney function, family history, and other risk-enhancing conditions.

Second, in patients with metabolic syndrome, type 2 diabetes, elevated triglycerides, obesity, chronic kidney disease, or clinical features suggesting LDL-C/apoB discordance, non-HDL-C should be reviewed carefully as a marker of atherogenic cholesterol burden. ApoB measurement adds precision where particle burden is uncertain, where LDL-C appears discordant with risk, or where treatment decisions are marginal.

Third, Lipoprotein(a) (Lp(a)) should be measured at least once in adulthood where feasible. Pragmatically, this can be done at the time of first formal cardiovascular risk assessment, in patients with premature ASCVD, in those with familial hypercholesterolaemia (FH), in people with strong family history, or when clinical risk appears disproportionate to standard lipid values. This does not necessarily require adding Lp(a) to every repeat lipid panel; it can be operationalised as a parallel once-in-adulthood measurement pathway.

Fourth, repeat testing should be purposeful. In stable low-risk adults not receiving lipid-lowering therapy, short-interval repeat lipid testing is rarely useful. In treated patients, monitoring should be linked to treatment initiation, dose changes, adherence assessment, target achievement, clinical-status changes, or evolving risk rather than performed reflexively.

Fifth, health systems should prioritise systematic lipid assessment in high-risk groups. These include patients with FH, diabetes, chronic kidney disease, established ASCVD, premature cardiovascular disease, strong family history, and communities with disproportionate cardiovascular risk. The problem in many systems is not too much testing overall, but the wrong distribution of testing: repeated panels in stable low-risk people and missed baseline assessment in those most likely to benefit.

Sixth, abnormal lipid results must be linked to structured action rather than recorded and set aside. Testing without risk calculation, communication, treatment access, or follow-up is incomplete care. Health systems should build pathways that connect lipid measurement to risk estimation, treatment decisions, medication access, adherence support, and recall. Practical mechanisms include electronic recall systems that flag untreated abnormal results, linked prescribing protocols that reduce the gap between an elevated LDL-C and a statin prescription, and practice-level audit that tracks whether high-risk patients have achieved guideline targets. Without these structural supports, expanded testing produces information without prevention.

10.3 Implications for Patients

Patients receiving lipid panel results should understand that a cholesterol test measures important blood biomarkers relevant to atherosclerotic cardiovascular and cerebrovascular disease risk, including coronary artery disease, ischaemic stroke, and peripheral artery disease. It is useful and often essential, but it is only one part of a larger cardiovascular risk picture.

A proper risk assessment also includes blood pressure, smoking or nicotine exposure, diabetes status, kidney function, age, sex, family history, medication use, and other clinical factors. A clinician needs these pieces together to estimate actual cardiovascular risk and decide whether lifestyle changes, medication, repeat testing, or additional evaluation are appropriate.

A result within the laboratory "normal" range does not mean cardiovascular risk is zero. This is especially true if other risk factors are present, such as high blood pressure, diabetes, smoking, chronic kidney disease, strong family history, or inherited risks such as Lp(a) or FH. Conversely, a result above the laboratory reference range does not always mean urgent treatment is required. The right response depends on absolute risk, lifetime risk, age, medical history, and whether the abnormality is mild, persistent, severe, or part of a broader pattern.

Most people who receive a standard lipid panel will not need additional specialist testing. For a small number of patients, however, further evaluation may be appropriate. This includes people with very high LDL-C, a family history of early heart disease, premature cardiovascular disease, or unexplained risk despite apparently acceptable cholesterol values. In those cases, additional testing may include assessment for FH, measurement of Lp(a), apoB testing, or imaging in selected circumstances. A clinician is best placed to judge whether any of these apply.

The practical message is simple: the lipid panel is a starting point for a conversation, not the end of the conversation.

10.4 Top 5 Unanswered Research and Policy Questions

1. What is the optimal approach to once-in-adulthood Lipoprotein(a) screening? Guidelines increasingly support at least one Lp(a) measurement in adulthood where feasible, but implementation remains unresolved. Key questions include the best age for testing, whether screening should occur in childhood, early adulthood, or at first formal cardiovascular risk assessment, and how results should be communicated when specific Lp(a)-lowering outcome-proven therapies are not yet widely available. The key evidence gap is no longer whether Lp(a) predicts risk, but whether direct pharmacological lowering of Lp(a) reduces hard cardiovascular outcomes at acceptable cost and scale. Pilot programmes, registry-based implementation studies, and updated health-economic models will be needed to guide policy.

2. In which populations would apoB-guided treatment improve outcomes beyond LDL-C- or non-HDL-C-guided strategies? Observational and discordance studies strongly support apoB as a better marker of atherogenic particle burden when LDL-C and particle number diverge. This is particularly relevant in diabetes, metabolic syndrome, obesity, hypertriglyceridaemia, chronic kidney disease, and very low achieved LDL-C states. However, no large-scale randomised clinical trial has directly compared an apoB-guided treatment strategy with an LDL-C- or non-HDL-C-guided strategy using hard cardiovascular endpoints. Future research should clarify whether apoB should become the primary treatment target in selected populations, whether non-HDL-C is sufficient in resource-limited settings, and how apoB thresholds should be implemented in routine care.

3. What is the impact of reframing cholesterol testing as a "baseline" rather than "standard" assessment? The hypothesis is plausible: baseline framing may reduce false reassurance, discourage unnecessary repeat testing, improve patient understanding, and encourage appropriate contextual assessment. However, direct evidence is lacking. Implementation science studies could test whether language changes affect patient comprehension, clinician ordering behaviour, risk-calculator use, statin initiation, repeat testing frequency, follow-up completion, and anxiety. Cluster-randomised primary-care studies or embedded electronic-health-record trials would be particularly useful.

4. How can lipid testing be sustainably integrated into primary care in low-resource settings? Point-of-care lipid analysers, non-fasting protocols, task-shifted risk assessment, and simplified treatment pathways offer potential solutions. However, comparative effectiveness evidence remains limited in real-world low- and middle-income settings. Key research needs include quality assurance for point-of-care testing, linkage of abnormal results to affordable statins, integration with hypertension and diabetes care, cost-effectiveness under constrained budgets, and strategies to reduce socioeconomic inequity in access. WHO PEN and HEARTS provide frameworks, but implementation evidence across diverse health systems remains incomplete.

5. How should cholesterol testing be embedded within emerging digital and AI-assisted cardiovascular risk platforms? Digital and AI-based cardiovascular risk tools are likely to expand rapidly. These systems may combine lipid data with blood pressure, glycaemia,

family history, imaging, genomics, social determinants, and electronic health-record data. They could improve risk prediction, but they also introduce new risks. Governance frameworks will be needed to ensure that algorithms are validated in diverse populations, calibrated where lipid data are missing, transparent in how they weight biomarkers, and safe for groups with limited access to laboratory testing. AI tools should not amplify inequity by treating absence of lipid data as absence of lipid risk, nor should they encourage over-reliance on algorithmic scores derived from incomplete or biased input data.

10.5 Final Statement

Routine cholesterol testing is neither obsolete nor sufficient.

It provides essential baseline information for cardiovascular risk assessment, dyslipidaemia diagnosis, treatment initiation, and monitoring. It has enabled therapies that have saved lives. Its scalability, affordability, and clinical familiarity make it one of the most important tools in preventive medicine.

But modern cardiovascular prevention requires precision about what the test can and cannot do. A standard lipid panel does not measure apoB, Lp(a), glycaemic status, blood pressure, family history, kidney function, inflammation, or subclinical plaque. In many patients, that does not matter. In selected patients, it matters profoundly.

The most defensible framing is therefore not that routine cholesterol testing should be replaced, but that it should be understood as a baseline lipid assessment: an indispensable first layer of cardiovascular risk evaluation, to be interpreted in context and expanded when evidence, risk, and resources justify doing so.

The task ahead is practical and global: make baseline lipid testing available where it is absent, use it purposefully where it is overused, interpret it more intelligently where it is misunderstood, and expand it selectively where modern evidence shows that the baseline panel is not enough.

References and Evidence Base

References are organised by topic rather than order of appearance. The bibliography includes major clinical guidelines, consensus statements, landmark epidemiological studies, randomized lipid-lowering trials, laboratory-method papers, global health policy documents, health-economic analyses, and selected risk-communication studies.

Major Guidelines, Consensus Statements, and Policy Frameworks

1–28 · 28 references

1. American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. 2026 ACC/AHA/AACVPR/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA Guideline on the Management of Dyslipidemia. *Circulation*. 2026. doi:10.1161/CIR.0000000000001423.
2. Grundy SM, Stone NJ, Bailey AL, et al. 2018 AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA Guideline on the Management of Blood Cholesterol. *Circulation*. 2019;139(25):e1082-e1143. doi:10.1161/CIR.0000000000000625.
3. Goff DC Jr, Lloyd-Jones DM, Bennett G, et al. 2013 ACC/AHA Guideline on the Assessment of Cardiovascular Risk. *Journal of the American College of Cardiology*. 2014;63(25 Pt B):2935-2959. doi:10.1016/j.jacc.2013.11.005.
4. Khan SS, Matsushita K, Sang Y, et al. Development and validation of the American Heart Association's PREVENT equations. *Circulation*. 2024;149(6):430-449. doi:10.1161/CIRCULATIONAHA.123.067626.
5. Mach F, Baigent C, Catapano AL, et al. 2019 ESC/EAS Guidelines for the management of dyslipidaemias: lipid modification to reduce cardiovascular risk. *European Heart Journal*. 2020;41(1):111-188. doi:10.1093/eurheartj/ehz455.
6. Mach F, Koskinas KC, Roeters van Lennep JE, et al. 2025 Focused Update of the 2019 ESC/EAS Guidelines for the management of dyslipidaemias. *European Heart Journal*. 2025;46(42):4359-4378. doi:10.1093/eurheartj/ehaf190.
7. National Institute for Health and Care Excellence. *Cardiovascular disease: risk assessment and reduction, including lipid modification*. NICE Guideline NG238. London: NICE; 2023.
8. US Preventive Services Task Force. Statin Use for the Primary Prevention of Cardiovascular Disease in Adults: US Preventive Services Task Force Recommendation Statement. *JAMA*. 2022;328(8):746-753. doi:10.1001/jama.2022.13044.
9. US Preventive Services Task Force. *Lipid Disorders in Adults (Cholesterol, Dyslipidemia): Screening*. Archived recommendation statement. Rockville, MD: USPSTF; 2008. Accessed May 2026.
10. Expert Panel on Detection, Evaluation, and Treatment of High Blood Cholesterol in Adults. Executive Summary of the Third Report of the National Cholesterol Education Program Expert Panel on Detection, Evaluation, and Treatment of High Blood Cholesterol in Adults. *JAMA*. 2001;285(19):2486-2497.
11. Pearson GJ, Thanassoulis G, Anderson TJ, et al. 2021 Canadian Cardiovascular Society Guidelines for the Management of Dyslipidemia for the Prevention of Cardiovascular Disease in Adults. *Canadian Journal of Cardiology*. 2021;37(8):1129-1150. doi:10.1016/j.cjca.2021.03.016.
12. Kolber MR, Korownyk C, Cauchon M, et al. PEER simplified lipid guideline 2023 update: prevention and management of cardiovascular disease in primary care. *Canadian Family Physician*. 2023;69(10):675-683.
13. National Heart Foundation of Australia. *Australian Guideline for Assessing and Managing Cardiovascular Disease Risk*. Melbourne: National Heart Foundation of Australia; 2023.

14. Li J, Zhao S, Zhao D, et al. 2023 Chinese guideline for lipid management. *Frontiers in Pharmacology*. 2023;14:1190934. doi:10.3389/fphar.2023.1190934.

EVIDENCE NOTE: Includes recommendation for lipid testing as a routine item in primary, middle, and high school entry physical examinations.

15. Klug E, Raal FJ, Marais AD, et al. South African dyslipidaemia guideline consensus statement. *South African Medical Journal*. 2012;102(3 Pt 2):178-187.
16. Gemeinsamer Bundesausschuss. *Richtlinie des Gemeinsamen Bundesausschusses über die Gesundheitsuntersuchungen zur Früherkennung von Krankheiten (Gesundheitsuntersuchungs-Richtlinie)*. Berlin: G-BA; 2019.

EVIDENCE NOTE: Statutory health check-up: once at age 18–34, every 3 years from age 35; lipid profile included from age 35.

17. World Health Organization. *Prevention of Cardiovascular Disease: Guidelines for Assessment and Management of Total Cardiovascular Risk*. Geneva: WHO; 2007.
18. World Health Organization. *HEARTS: Technical Package for Cardiovascular Disease Management in Primary Health Care: Risk-Based CVD Management*. Geneva: WHO; 2020.
19. World Health Organization. *Package of Essential Noncommunicable Disease Interventions for Primary Health Care in Low-Resource Settings*. Geneva: WHO; 2020.
20. WHO CVD Risk Chart Working Group. World Health Organization cardiovascular disease risk charts: revised models to estimate risk in 21 global regions. *Lancet Global Health*. 2019;7(10):e1332-e1345. doi:10.1016/S2214-109X(19)30318-3.
21. World Health Organization. *Global Action Plan for the Prevention and Control of Noncommunicable Diseases 2013-2020*. Geneva: WHO; 2013.
22. Kronenberg F, Bedlington N, Ademi Z, et al. The Brussels International Declaration on Lipoprotein(a) Testing and Management. *Atherosclerosis*. 2025;406:119218. doi:10.1016/j.atherosclerosis.2025.119218.
23. Ray KK, Ference BA, Séverin T, et al. World Heart Federation Cholesterol Roadmap 2022. *Global Heart*. 2022;17(1):75. doi:10.5334/gh.1154.
24. Soffer DE, Marston NA, Maki KC, et al. Role of apolipoprotein B in the clinical management of cardiovascular risk in adults: an expert clinical consensus from the National Lipid Association. *Journal of Clinical Lipidology*. 2024. doi:10.1016/j.jacl.2024.08.013.
25. Kronenberg F, Mora S, Stroes ESG, et al. Lipoprotein(a) in atherosclerotic cardiovascular disease and aortic stenosis: a European Atherosclerosis Society consensus statement. *European Heart Journal*. 2022;43(39):3925-3946. doi:10.1093/eurheartj/ehac361.
26. Reyes-Soffer G, Ginsberg HN, Berglund L, et al. Lipoprotein(a): a genetically determined, causal, and prevalent risk factor for atherosclerotic cardiovascular disease. *Arteriosclerosis, Thrombosis, and Vascular Biology*. 2022;42(1):e48-e60. doi:10.1161/ATV.000000000000147.
27. Wilson DP, Jacobson TA, Jones PH, et al. Use of lipoprotein(a) in clinical practice: a biomarker whose time has come. A scientific statement from the National Lipid Association. *Journal of Clinical Lipidology*. 2019;13(3):374-392. doi:10.1016/j.jacl.2019.04.010.
28. Kotseva K, De Backer G, De Bacquer D, et al. Lifestyle and impact on cardiovascular risk factor control in coronary patients across 27 countries: results from the European Society of Cardiology ESC-EORP EUROASPIRE V registry. *European Journal of Preventive Cardiology*. 2019;26(8):824-835. doi:10.1177/2047487318825350.

EVIDENCE NOTE: EUROASPIRE V informs real-world treatment and risk-factor control gaps in coronary patients.

Laboratory Methods, Lipid Measurement, and Reporting 29–43 · 15 references

29. McNamara JR, Warnick GR, Cooper GR. A brief history of lipid and lipoprotein measurements and their contribution to clinical chemistry. *Clinica Chimica Acta*. 2006;369(2):158-167. doi:10.1016/j.cca.2006.02.041.
30. Copeland BE. Serum cholesterol methodology: 100 years of development. *Annals of Clinical and Laboratory Science*. 1990;20(1):1-11.
31. Friedewald WT, Levy RI, Fredrickson DS. Estimation of the concentration of low-density lipoprotein cholesterol in plasma, without use of the preparative ultracentrifuge. *Clinical Chemistry*. 1972;18(6):499-502.
32. Abell LL, Levy BB, Brodie BB, Kendall FE. A simplified method for the estimation of total cholesterol in serum and demonstration of its specificity. *Journal of Biological Chemistry*. 1952;195(1):357-366.
33. Schoenheimer R, Sperry WM. A micromethod for the determination of free and combined cholesterol. *Journal of Biological Chemistry*. 1934;106:745-760.
34. Fredrickson DS, Levy RI, Lees RS. Fat transport in lipoproteins: an integrated approach to mechanisms and disorders. *New England Journal of Medicine*. 1967;276:34-42, 94-103, 148-156, 215-225, 273-281.
35. Martin SS, Blaha MJ, Elshazly MB, et al. Comparison of a novel method vs the Friedewald equation for estimating low-density lipoprotein cholesterol levels from the standard lipid panel. *JAMA*. 2013;310(19):2061-2068. doi:10.1001/jama.2013.280532.
36. Sampson M, Ling C, Sun Q, et al. A new equation for calculation of low-density lipoprotein cholesterol in patients with normolipidemia and/or hypertriglyceridemia. *JAMA Cardiology*. 2020;5(5):540-548. doi:10.1001/jamacardio.2020.0013.
37. Nordestgaard BG, Langsted A, Mora S, et al. Fasting is not routinely required for determination of a lipid profile: clinical and laboratory implications. *European Heart Journal*. 2016;37(25):1944-1958. doi:10.1093/eurheartj/ehw152.
38. Langsted A, Freiberg JJ, Nordestgaard BG. Fasting and nonfasting lipid levels: influence of normal food intake on lipids, lipoproteins, apolipoproteins, and cardiovascular risk prediction. *Circulation*. 2008;118(20):2047-2056. doi:10.1161/CIRCULATIONAHA.108.804146.
39. Cao J, Donato LJ, El-Khoury JM, Goldberg AC, Meeusen JW, Remaley AT. ADLM Guidance Document on the Measurement and Reporting of Lipids and Lipoproteins. *Journal of Applied Laboratory Medicine*. 2024;9(5):1040-1056. doi:10.1093/jalm/jfae057.
40. Wilson PWF, Jacobson TA, Martin SS, et al. Lipid measurements in the management of cardiovascular diseases: practical recommendations from a National Lipid Association scientific statement. *Journal of Clinical Lipidology*. 2021. doi:10.1016/j.jacl.2021.09.046.
41. Langlois MR, Chapman MJ, Cobbaert C, et al. Quantifying atherogenic lipoproteins: current and future challenges in the era of personalized medicine and very low concentrations of LDL cholesterol. *Clinical Chemistry*. 2018;64(7):1006-1033. doi:10.1373/clinchem.2018.287037.
42. Nordestgaard BG, Langlois MR, Langsted A, et al. Quantifying atherogenic lipoproteins for lipid-lowering strategies: consensus-based recommendations from EAS and EFLM. *Atherosclerosis*. 2020;294:76-85. doi:10.1016/j.atherosclerosis.2019.12.005.
43. Remaley AT, Dayspring TD, Warnick GR. Lipids, lipoproteins, apolipoproteins, and other cardiovascular risk factors. In: Feingold KR, Anawalt B, Blackman MR, et al., editors. *Endotext*. South Dartmouth, MA: [MDText.com](https://www.mdtext.com/); updated periodically.

Historical Development and Epidemiology 44–54 · 11 references

44. Kannel WB, Dawber TR, Kagan A, Revotskie N, Stokes J. Factors of risk in the development of coronary heart disease: six-year follow-up experience. The Framingham Study. *Annals of Internal Medicine*. 1961;55:33-50.
45. Keys A, Aravanis C, Blackburn HW, Van Buchem FS, Buzina R, Djordjević BD, et al. Epidemiological studies related to coronary heart disease: characteristics of men aged 40-59 in seven countries. *Acta Medica Scandinavica Supplementum*. 1966;460:1-392.

46. Menotti A, Puddu PE. The lipid-heart hypothesis from the Seven Countries Study. *Current Opinion in Lipidology*. 2025;36(4):191-197.
47. Yusuf S, Hawken S, Ounpuu S, et al. Effect of potentially modifiable risk factors associated with myocardial infarction in 52 countries: the INTERHEART study. *Lancet*. 2004;364(9438):937-952. doi:10.1016/S0140-6736(04)17018-9.
48. McQueen MJ, Hawken S, Wang X, et al. Lipids, lipoproteins, and apolipoproteins as risk markers of myocardial infarction in 52 countries: the INTERHEART study. *Lancet*. 2008;372(9634):224-233. doi:10.1016/S0140-6736(08)61076-4.
49. Emerging Risk Factors Collaboration. Major lipids, apolipoproteins, and risk of vascular disease. *JAMA*. 2009;302(18):1993-2000. doi:10.1001/jama.2009.1619.
50. NCD Risk Factor Collaboration. Repositioning of the global epicentre of non-optimal cholesterol. *Nature*. 2020;582:73-77. doi:10.1038/s41586-020-2338-1.
51. Roth GA, Mensah GA, Johnson CO, et al. Global burden of cardiovascular diseases and risk factors, 1990-2019: update from the GBD 2019 study. *Journal of the American College of Cardiology*. 2020;76(25):2982-3021. doi:10.1016/j.jacc.2020.11.010.
52. Yusuf S, Joseph P, Rangarajan S, et al. Modifiable risk factors, cardiovascular disease, and mortality in 155,722 individuals from 21 high-income, middle-income, and low-income countries: PURE. *Lancet*. 2020;395(10226):795-808. doi:10.1016/S0140-6736(19)32008-2.
53. Pencina MJ, D'Agostino RB Sr, Larson MG, Massaro JM, Vasan RS. Predicting the 30-year risk of cardiovascular disease: the Framingham Heart Study. *Circulation*. 2009;119(24):3078-3084. doi:10.1161/CIRCULATIONAHA.108.816694.
54. Martin SS, Niles J, Kaufman HW, et al. Lipid distributions in the Global Diagnostics Network across five continents. *European Heart Journal*. 2023;44(25):2305-2318. doi:10.1093/eurheartj/ehad371.

LDL-C Lowering Trials and Outcome Evidence 55-73 · 19 references

55. Lipid Research Clinics Program. The Lipid Research Clinics Coronary Primary Prevention Trial results. I. Reduction in incidence of coronary heart disease. *JAMA*. 1984;251(3):351-364.
56. Scandinavian Simvastatin Survival Study Group. Randomised trial of cholesterol lowering in 4444 patients with coronary heart disease: the Scandinavian Simvastatin Survival Study. *Lancet*. 1994;344(8934):1383-1389.
57. Shepherd J, Cobbe SM, Ford I, et al. Prevention of coronary heart disease with pravastatin in men with hypercholesterolemia. *New England Journal of Medicine*. 1995;333(20):1301-1307.
58. Sacks FM, Pfeffer MA, Moya LA, et al. The effect of pravastatin on coronary events after myocardial infarction in patients with average cholesterol levels. *New England Journal of Medicine*. 1996;335(14):1001-1009.
59. The Long-Term Intervention with Pravastatin in Ischaemic Disease Study Group. Prevention of cardiovascular events and death with pravastatin in patients with coronary heart disease and a broad range of initial cholesterol levels. *New England Journal of Medicine*. 1998;339(19):1349-1357.
60. Heart Protection Study Collaborative Group. MRC/BHF Heart Protection Study of cholesterol lowering with simvastatin in 20,536 high-risk individuals. *Lancet*. 2002;360(9326):7-22.
61. Cholesterol Treatment Trialists' Collaborators. Efficacy and safety of cholesterol-lowering treatment: prospective meta-analysis of data from 90,056 participants in 14 randomized trials of statins. *Lancet*. 2005;366(9493):1267-1278.
62. Cholesterol Treatment Trialists' Collaboration. Efficacy and safety of more intensive lowering of LDL cholesterol: meta-analysis of data from 170,000 participants in 26 randomized trials. *Lancet*. 2010;376(9753):1670-1681.
63. Cholesterol Treatment Trialists' Collaboration. The effects of lowering LDL cholesterol with statin therapy in people at low risk of vascular disease: meta-analysis of individual data from 27 randomized trials. *Lancet*. 2012;380(9841):581-590.

64. Cholesterol Treatment Trialists' Collaboration. Efficacy and safety of statin therapy in older people: a meta-analysis of individual participant data from 28 randomized controlled trials. *Lancet*. 2019;393(10170):407-415.
65. Cannon CP, Blazing MA, Giugliano RP, et al. Ezetimibe added to statin therapy after acute coronary syndromes. *New England Journal of Medicine*. 2015;372(25):2387-2397.
66. Sabatine MS, Giugliano RP, Keech AC, et al. Evolocumab and clinical outcomes in patients with cardiovascular disease. *New England Journal of Medicine*. 2017;376(18):1713-1722.
67. Schwartz GG, Steg PG, Szarek M, et al. Alirocumab and cardiovascular outcomes after acute coronary syndrome. *New England Journal of Medicine*. 2018;379(22):2097-2107.
68. Nissen SE, Lincoff AM, Brennan D, et al. Bempedoic acid and cardiovascular outcomes in statin-intolerant patients. *New England Journal of Medicine*. 2023;388(15):1353-1364.
69. Ray KK, Wright RS, Kallend D, et al. Two phase 3 trials of inclisiran in patients with elevated LDL cholesterol. *New England Journal of Medicine*. 2020;382(16):1507-1519.
70. Ridker PM, Danielson E, Fonseca FAH, et al. Rosuvastatin to prevent vascular events in men and women with elevated C-reactive protein. *New England Journal of Medicine*. 2008;359(21):2195-2207.
71. Ridker PM, Everett BM, Thuren T, et al. Antiinflammatory therapy with canakinumab for atherosclerotic disease. *New England Journal of Medicine*. 2017;377(12):1119-1131.
72. Bhatt DL, Steg PG, Miller M, et al. Cardiovascular risk reduction with icosapent ethyl for hypertriglyceridemia. *New England Journal of Medicine*. 2019;380(1):11-22.
73. Collins R, Reith C, Emberson J, et al. Interpretation of the evidence for the efficacy and safety of statin therapy. *Lancet*. 2016;388(10059):2532-2561.

Causal Evidence and Mendelian Randomisation 74-77 · 4 references

74. Ference BA, Ginsberg HN, Graham I, et al. Low-density lipoproteins cause atherosclerotic cardiovascular disease. 1. Evidence from genetic, epidemiologic, and clinical studies. *European Heart Journal*. 2017;38(32):2459-2472.
75. Borén J, Chapman MJ, Krauss RM, et al. Low-density lipoproteins cause atherosclerotic cardiovascular disease: pathophysiological, genetic, and therapeutic insights. *European Heart Journal*. 2020;41(24):2313-2330.
76. Ference BA, Bhatt DL, Catapano AL, et al. Association of genetic variants related to combined exposure to lower LDL-C and lower systolic blood pressure with lifetime risk of cardiovascular disease. *JAMA*. 2019;322(14):1381-1391.
77. Holmes MV, Ala-Korpela M, Smith GD. Mendelian randomization in cardiometabolic disease: challenges in evaluating causality. *Nature Reviews Cardiology*. 2017;14(10):577-590.

ApoB, Non-HDL-C, Remnant Cholesterol, and Discordance

78-88 · 11 references

78. Sniderman AD, Williams K, Contois JH, et al. A meta-analysis of low-density lipoprotein cholesterol, non-HDL cholesterol, and apolipoprotein B as markers of cardiovascular risk. *Circulation: Cardiovascular Quality and Outcomes*. 2011;4(3):337-345.
79. Sniderman AD, Thanassoulis G, Glavinovic T, et al. Apolipoprotein B particles and cardiovascular disease: a narrative review. *JAMA Cardiology*. 2019;4(12):1287-1295.
80. Boekholdt SM, Arsenault BJ, Mora S, et al. Association of LDL cholesterol, non-HDL cholesterol, and apolipoprotein B levels with risk of cardiovascular events among patients treated with statins. *JAMA*. 2012;307(12):1302-1309.
81. Sayed A, Peterson ED, Virani SS, Sniderman AD, Navar AM. Individual variation in the distribution of apolipoprotein B levels across the spectrum of LDL-C or non-HDL-C levels. *JAMA Cardiology*. 2024. doi:10.1001/jamacardio.2024.1310.

82. Sehayek D, Cole J, Björnson E, et al. ApoB, LDL-C, and non-HDL-C as markers of cardiovascular risk. *Journal of Clinical Lipidology*. 2025. doi:10.1016/j.jacl.2025.05.024.
83. Johannesen CD, Mortensen MB, Nordestgaard BG, Langsted A. Discordance analyses comparing LDL cholesterol, non-HDL cholesterol, and apolipoprotein B for cardiovascular risk estimation. *Atherosclerosis*. 2025;403:119139. doi:10.1016/j.atherosclerosis.2025.119139.
84. Wolska A, Lloyd-Jones DM, Remaley AT. Measure Apolipoprotein B — If We Believe What We Say About Precision Medicine. *Circulation*. 2025;151(5):257-259. doi:10.1161/CIRCULATIONAHA.123.067559. [Demonstrates that >25% of individuals have discordantly higher apoB than LDL-C or non-HDL-C.]
85. Brunner FJ, Waldeyer C, Ojeda F, et al. Application of non-HDL cholesterol for population-based cardiovascular risk stratification. *Lancet*. 2019;394(10215):2173-2183.
86. Nordestgaard BG. Triglyceride-rich lipoproteins and atherosclerotic cardiovascular disease: new insights from epidemiology, genetics, and biology. *Circulation Research*. 2016;118(4):547-563.
87. Varbo A, Benn M, Tybjaerg-Hansen A, Nordestgaard BG. Remnant cholesterol as a causal risk factor for ischemic heart disease. *Journal of the American College of Cardiology*. 2013;61(4):427-436.
88. Yang X, Zhang B, Cheng Y, Fu S, Jin H. Association of remnant cholesterol with risk of cardiovascular disease events, stroke, and mortality: a systematic review and meta-analysis. *Atherosclerosis*. 2023;371:21-31.

Lipoprotein(a) 89–100 · 12 references

89. Nordestgaard BG, Chapman MJ, Ray K, et al. Lipoprotein(a) as a cardiovascular risk factor: current status. *European Heart Journal*. 2010;31(23):2844-2853.
90. Kamstrup PR, Tybjaerg-Hansen A, Steffensen R, Nordestgaard BG. Genetically elevated lipoprotein(a) and increased risk of myocardial infarction. *JAMA*. 2009;301(22):2331-2339.
91. Clarke R, Peden JF, Hopewell JC, et al. Genetic variants associated with Lp(a) lipoprotein level and coronary disease. *New England Journal of Medicine*. 2009;361(26):2518-2528.
92. Burgess S, Ference BA, Staley JR, et al. Association of LPA variants with risk of coronary disease and the implications for lipoprotein(a)-lowering therapies. *JAMA Cardiology*. 2018;3(7):619-627.
93. O'Donoghue ML, Fazio S, Giugliano RP, et al. Lipoprotein(a), PCSK9 inhibition, and cardiovascular risk. *Circulation*. 2019;139(12):1483-1492.
94. Bittner VA, Szarek M, Aylward PE, et al. Effect of alirocumab on lipoprotein(a) and cardiovascular events after acute coronary syndrome. *Journal of the American College of Cardiology*. 2020;75(2):133-144.
95. Tsimikas S, Karwatowska-Prokopczuk E, Gouni-Berthold I, et al. Lipoprotein(a) reduction in persons with cardiovascular disease. *New England Journal of Medicine*. 2020;382(3):244-255.
96. O'Donoghue ML, Rosenson RS, Gencer B, et al. Small interfering RNA to reduce lipoprotein(a) in cardiovascular disease. *New England Journal of Medicine*. 2022;387(20):1855-1864.
97. Nissen SE, Wolski K, Balog C, et al. Single ascending dose study of a short interfering RNA targeting lipoprotein(a) production. *JAMA*. 2022;327(17):1679-1687.
98. Nicholls SJ, Nissen SE, Fleming C, et al. Muvalaplin, an oral small molecule inhibitor of lipoprotein(a) formation: a randomized clinical trial. *JAMA*. 2023;330(11):1042-1053.
99. Bhatia HS, Hurst S, Desai P, Zhu W, Yeang C. Lipoprotein(a) testing trends in a large academic health system in the United States. *Journal of the American Heart Association*. 2023;12:e031255. doi:10.1161/JAHA.123.031255.
100. Zafir B, Aker A, Saliba W. Lipoprotein(a) testing in clinical practice: real-life data from a large health-care provider. *European Journal of Preventive Cardiology*. 2022. doi:10.1093/eurjpc/zwac124.

Coronary Artery Calcium, Imaging, and Risk Reclassification

- .01. Greenland P, Blaha MJ, Budoff MJ, Erbel R, Watson KE. Coronary calcium score and cardiovascular risk. *Journal of the American College of Cardiology*. 2018;72(4):434-447.
- .02. Blaha MJ, Cainzos-Achirica M, Greenland P, et al. Role of coronary artery calcium score of zero and other negative risk markers for cardiovascular disease. *Circulation*. 2016;133(9):849-858.
- .03. Yeboah J, Young R, McClelland RL, et al. Utility of nontraditional risk markers in atherosclerotic cardiovascular disease risk assessment. *Journal of the American College of Cardiology*. 2016;67(2):139-147.
- .04. Pletcher MJ, Pignone M, Earnshaw S, et al. Using the coronary artery calcium score to guide statin therapy: a cost-effectiveness analysis. *Circulation: Cardiovascular Quality and Outcomes*. 2014;7(2):276-284.

Global Access, Screening Practice, Overuse, Underuse, and Health Systems

105–114 · 10 references

- .05. Ochmann S, von Polenz I, et al. Diagnostic testing for hypertension, diabetes, and hypercholesterolaemia in low-income and middle-income countries: a cross-sectional study of data from 57 nationally representative surveys. *Lancet Global Health*. 2023;11:e1363-e1371. doi:10.1016/S2214-109X(23)00280-2.

EVIDENCE NOTE: Among adults meeting WHO PEN criteria for diagnostic cholesterol testing, only 39.7% had ever had their cholesterol measured; testing was associated with higher wealth and education.

- .06. O'Sullivan JW, Stevens S, Oke J, et al. The rise of cholesterol testing: how much is unnecessary? *British Journal of General Practice*. 2011;61(583):e81-e88. doi:10.3399/bjgp11X556227.
- .07. Robson J, Dostal I, Sheikh A, et al. The NHS Health Check in England: an evaluation of the first 4 years. *BMJ Open*. 2016;6(1):e008840.
- .08. Forster AS, Burgess C, Dodhia H, Fuller F, Miller J, McDermott L. Do health checks improve risk factor detection in primary care? Matched cohort study using electronic health records. *Journal of Public Health*. 2015;37(4):666-674.
- .09. Carrillo-Larco RM, Benites-Moya C, Anza-Ramírez C, et al. A systematic review of population-based studies on lipid profiles in Latin America and the Caribbean. *eLife*. 2020;9:e57980. doi:10.7554/eLife.57980.
- .10. Al Kuwari MG, et al. Assessing the impact of annual health screenings in identifying noncommunicable disease risk factors within Qatar's primary health care corporation Qatari registered population. *Frontiers in Public Health*. 2024;12:1305636.
- .11. Kotseva K, De Bacquer D, Jennings C, et al. Time trends in lifestyle and risk factor control in coronary patients: results from the EUROASPIRE V survey. *European Journal of Preventive Cardiology*. 2022;29(5):747-757.

EVIDENCE NOTE: EUROASPIRE V documents real-world lipid goal attainment in secondary prevention.

- .12. Millett ERC, Peters SAE, Woodward M. Sex differences in risk factors for myocardial infarction: cohort study of UK Biobank participants. *BMJ*. 2018;363:k4247.
- .13. Kengne AP, Ntyintyane L, Mayosi BM. Dyslipidaemia in South Africa: a systematic review of prevalence, determinants, and management. *South African Medical Journal*. 2016;106(6):589-594.

EVIDENCE NOTE: Provides region-specific lipid data for sub-Saharan Africa, supporting the South African content in Section 5.

- .14. Nordestgaard BG, Chapman MJ, Humphries SE, et al.** Familial hypercholesterolaemia is underdiagnosed and undertreated in the general population: guidance for clinicians to prevent coronary heart disease. *European Heart Journal*. 2013;34(45):3478-3490. doi:10.1093/eurheartj/eh273.

EVIDENCE NOTE: Landmark consensus on FH prevalence (~1 in 200–250) and the extent of underdiagnosis, underpinning Sections 3, 5, and 7.

Cost-Effectiveness and Implementation 115–119 · 5 references

- .15.** Pandya A, Sy S, Cho S, Weinstein MC, Gaziano TA. Cost-effectiveness of 10-year risk thresholds for initiation of statin therapy for primary prevention of cardiovascular disease. *JAMA*. 2015;314(2):142-150.
- .16.** Kerr M, Pears R, Miedzybrodzka Z, et al. Cost effectiveness of cascade testing for familial hypercholesterolaemia. *European Heart Journal*. 2017;38(23):1832-1839.
- .17.** Watts GF, Gidding S, Hegele RA, et al. International Atherosclerosis Society guidance for implementing best practice in the care of familial hypercholesterolaemia. *Nature Reviews Cardiology*. 2023;20:845-869. doi:10.1038/s41569-023-00892-0.
- .18.** Kazi DS, Moran AE, Coxson PG, et al. Cost-effectiveness of PCSK9 inhibitor therapy in patients with heterozygous familial hypercholesterolemia or atherosclerotic cardiovascular disease. *JAMA*. 2016;316(7):743-753.
- .19.** Morton JI, Kronenberg F, Daccord M, et al; Lp(a) International Taskforce Initiative. Lp(a) testing for the primary prevention of cardiovascular disease in high-income countries: a cost-effectiveness analysis. *Atherosclerosis*. 2025;409:120447. doi:10.1016/j.atherosclerosis.2025.120447.

Risk Communication, Patient Understanding, and Testing Behaviour

120–126 · 7 references

- .20.** Murdoch M, Wilt TJ. Cholesterol awareness after case-finding: do patients really know their cholesterol numbers? *American Journal of Preventive Medicine*. 1997;13(4):284-289.
- .21.** Goldman RE, Parker DR, Eaton CB, Borkan JM, Gramling R, Cover RT, Ahern DK. Patients' perceptions of cholesterol, cardiovascular disease risk, and risk communication strategies. *Annals of Family Medicine*. 2006;4(3):205-212.
- .22.** Krones T, Keller H, Sönnichsen A, et al. Absolute cardiovascular disease risk and shared decision making in primary care: a randomized controlled trial. *Annals of Family Medicine*. 2008;6(3):218-227.
- .23.** Bonner C, Jansen J, Newell BR, et al. Is the "heart age" concept helpful or harmful compared with absolute cardiovascular disease risk? *Medical Decision Making*. 2015;35(8):967-978.
- .24.** Hagger MS, Hardcastle SJ, Hu M, et al. Effects of medication, cardiovascular risk information, and health-risk label on risk perceptions and intentions in people with hypercholesterolaemia: a randomized controlled trial. *Annals of Behavioral Medicine*. 2019;53(2):158-168.
- .25.** Navar AM, Wang TY, Li S, et al. Patient-perceived versus actual risk of cardiovascular disease and associated willingness to consider statin therapy. *Circulation: Cardiovascular Quality and Outcomes*. 2017;10(5):e003253.
- .26.** Gigerenzer G, Gaissmaier W, Kurz-Milcke E, Schwartz LM, Woloshin S. Helping doctors and patients make sense of health statistics. *Psychological Science in the Public Interest*. 2007;8(2):53-96.

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HEARTFIRST BY SHYNTESY

Use the baseline more precisely

Cholesterol testing remains one of cardiovascular prevention's most important foundations. The task now is to use it more precisely: as a baseline assessment, interpreted in context, and expanded when evidence and risk justify doing so.

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